

Cell Systems, Volume 17

Supplemental information

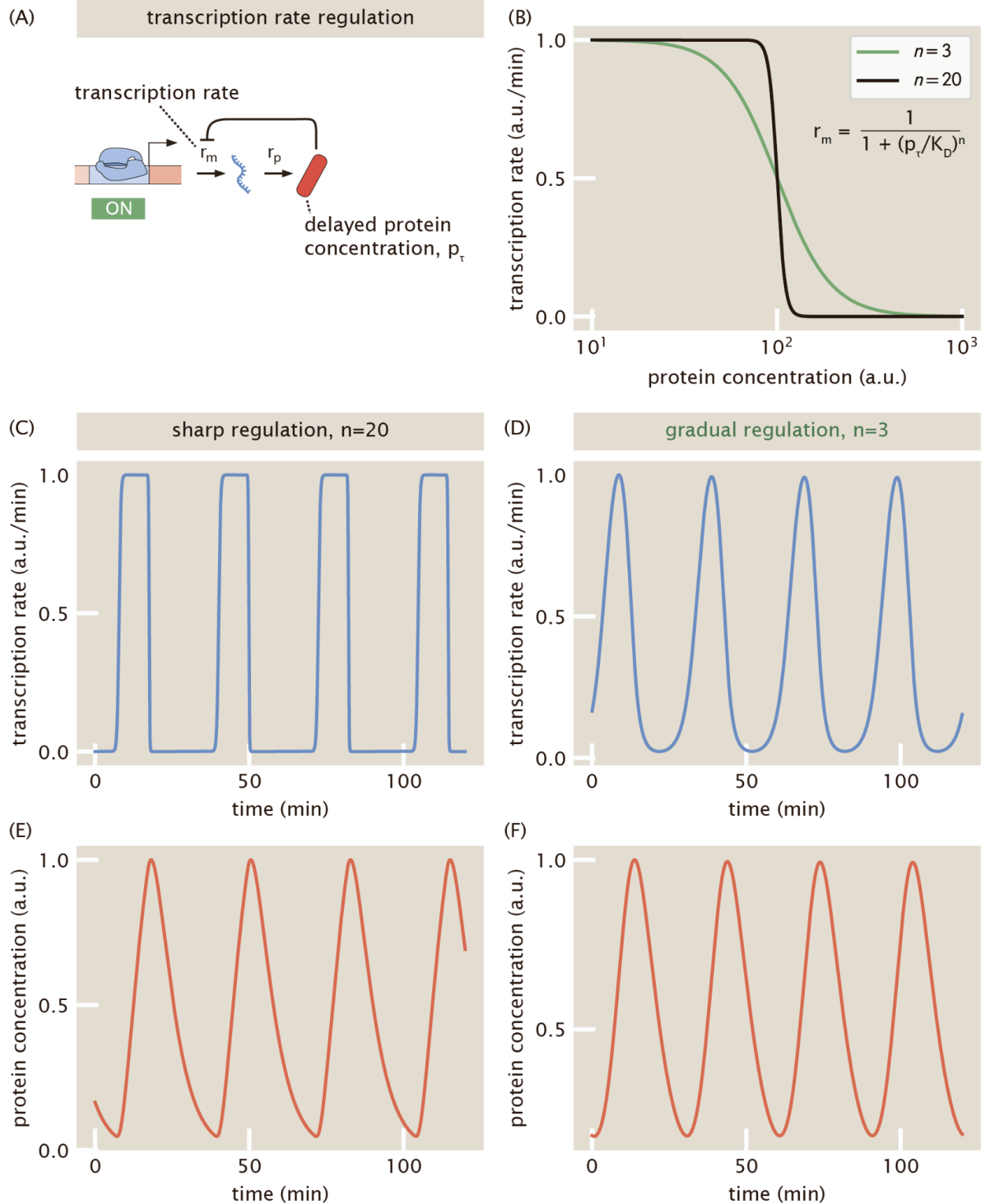
Single-cell transcriptional dynamics in a living vertebrate

Elizabeth Eck, Bruno Moretti, Brandon H. Schlomann, Jordão Bragantini, Merlin Lange, Xiang Zhao, Shruthi VijayKumar, Guillaume Valentin, Cristina Loureiro, Pablo Perez Franco, Chloé Jollivet, Virginie Braman, Baldemar Motomochi, Loïc A. Royer, Andrew C. Oates, and Hernan G. Garcia

Supporting Information

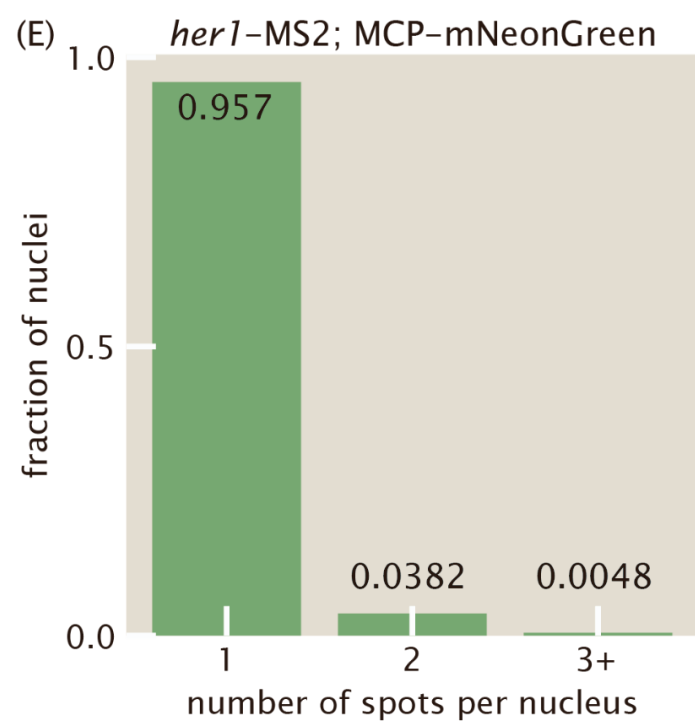
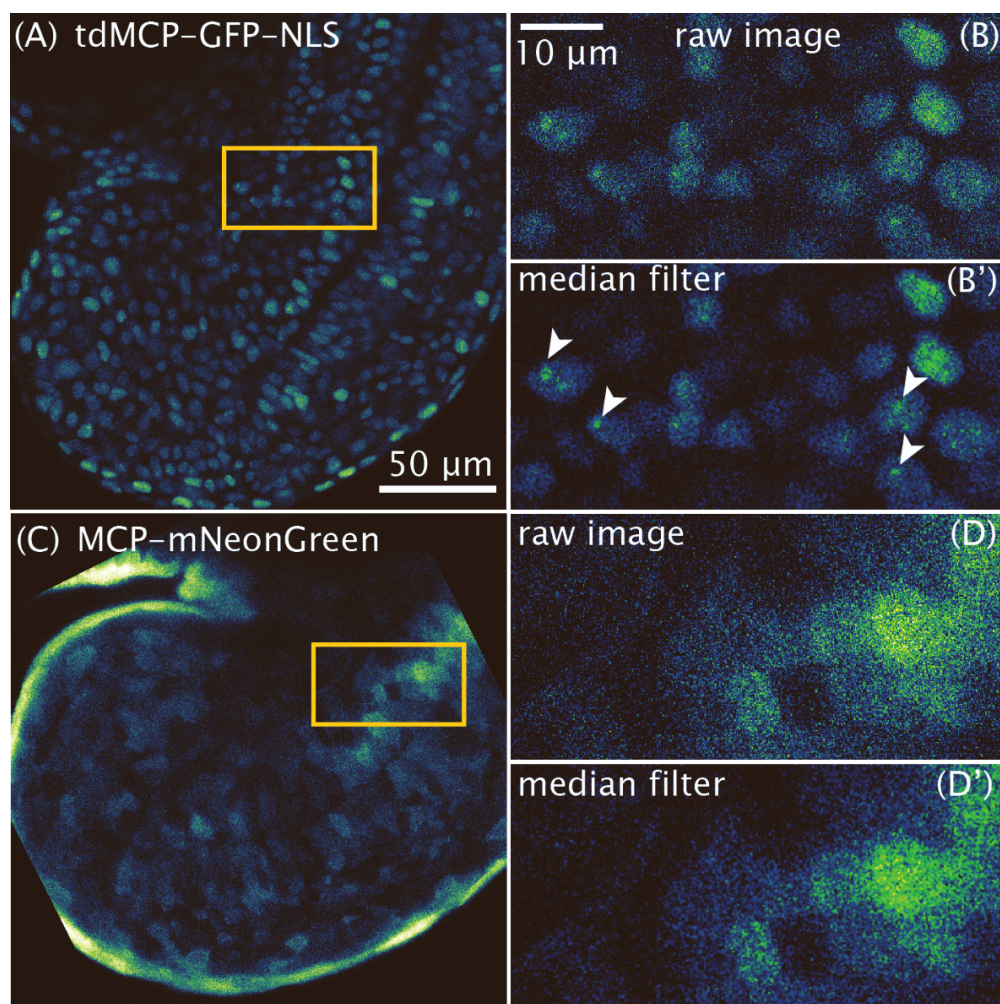
Supplemental Figures 1-21

Methods S1-S3

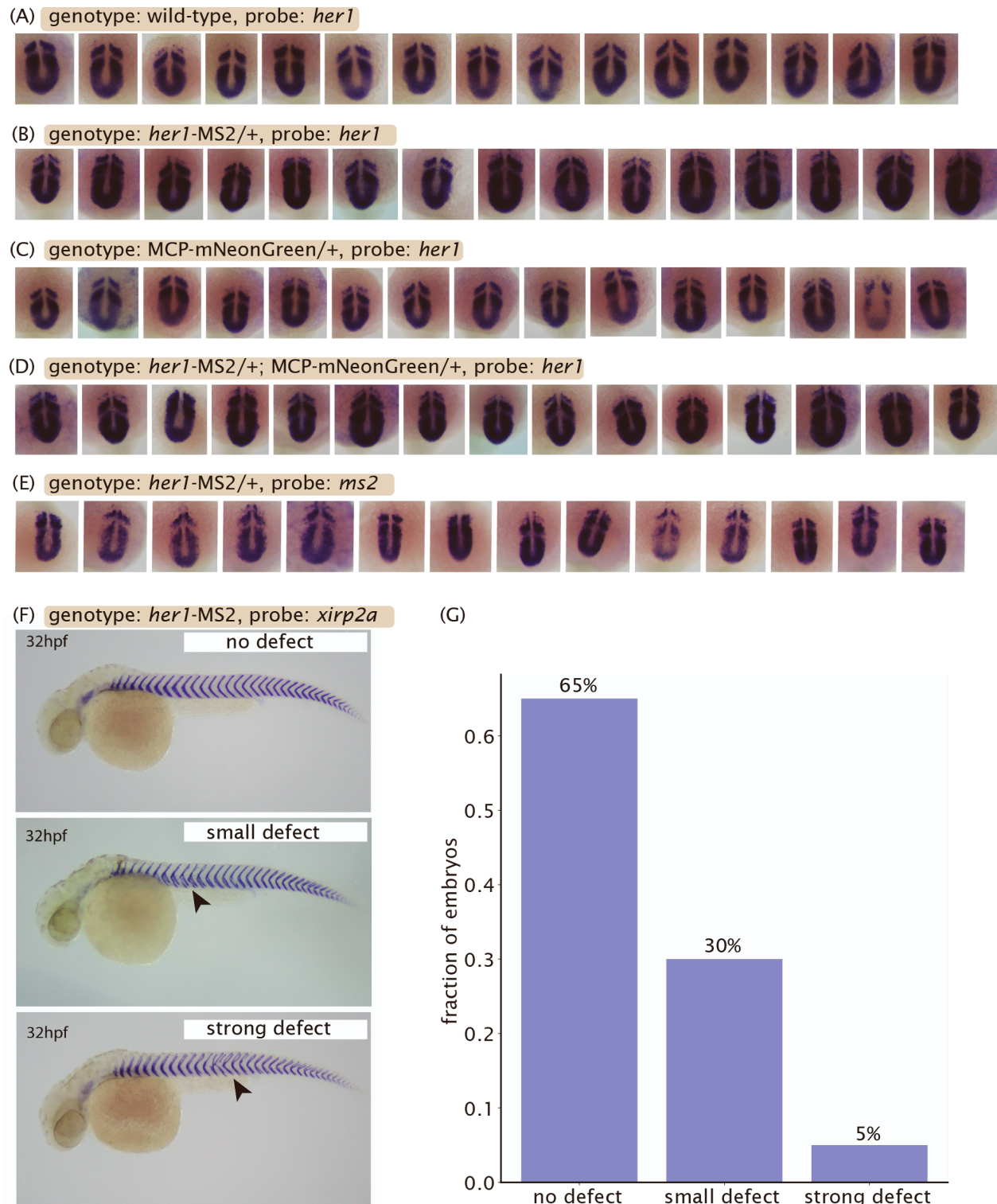


Supplemental Figure 1: A mechanistic example of how distinct *her1* transcriptional dynamics can give rise to similar protein dynamics. (A) Schematic of a deterministic model in which Her1 protein reduces the *her1* transcription rate. The model is given by the delay differential equations (48)-(49) in the Appendix, and the notation p_t represents $p(t-\tau)$, with τ

being the delay between transcription initiation and the appearance of mature mRNA in the cytoplasm. Not shown are mRNA and protein degradation. (B) Examples of gradual vs. sharp input-output functions for the repression, which are given by Hill functions with different Hill coefficients, n . (C) With a high Hill coefficient, the transcription rate switches over time in a square wave-like manner. (D) With a low Hill coefficient, the transcription rate oscillates in a sine wave-like manner. (E)-(F) Both situations produce smooth protein oscillations. Parameters: mRNA and protein degradation rates are both set to 0.23 1/min, $r_m = 10$ mRNAs/min, $r_p = 4.5$ proteins/mRNA/min, $K_D = 100$ proteins. For plotting, transcription rate and protein traces were normalized to their maximum.

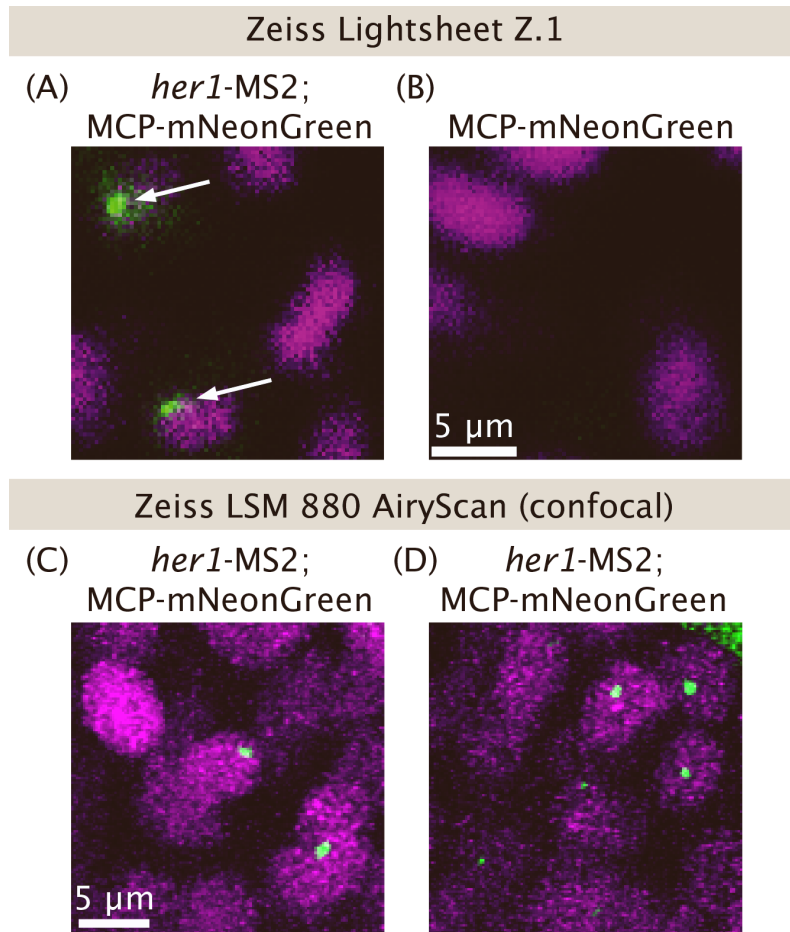


Supplemental Figure 2: A redesigned MCP line shows reduced aggregation. (A)-(B) Single z-slice of an embryo carrying tdMCP-GFP-NLS from ²⁰ showing nuclear localization and aggregation, which is highlighted in the inset in (B). We also show a median-filtered image of the inset for clarity. (C)-(D) Single z-slice of an embryo carrying MCP-mNeonGreen, showing no aggregation. (E) Quantification of number of fluorescent spots per nucleus detected in a heterozygous *her1*-MS2 embryo; MCP-mNeonGreen animal, which shows that approximately 95% of nuclei have only a single spot corresponding to the site of nascent transcription at the reporter. Multiple spots per nucleus most likely reflect false positive detections, but may also represent cells that have replicated their genome prior to division, and thus have two transcriptional spots. This analysis was done on $n=7847$ observations of 403 nuclei pooled across 163 time points, with repeated observations of the same nucleus counting as a separate observation. However, we obtain the same result if we restrict our analysis to the single time point with the highest number of detected spots (fraction of nuclei with one spot = 0.95). Nuclei with no detected spots were not included in this analysis.



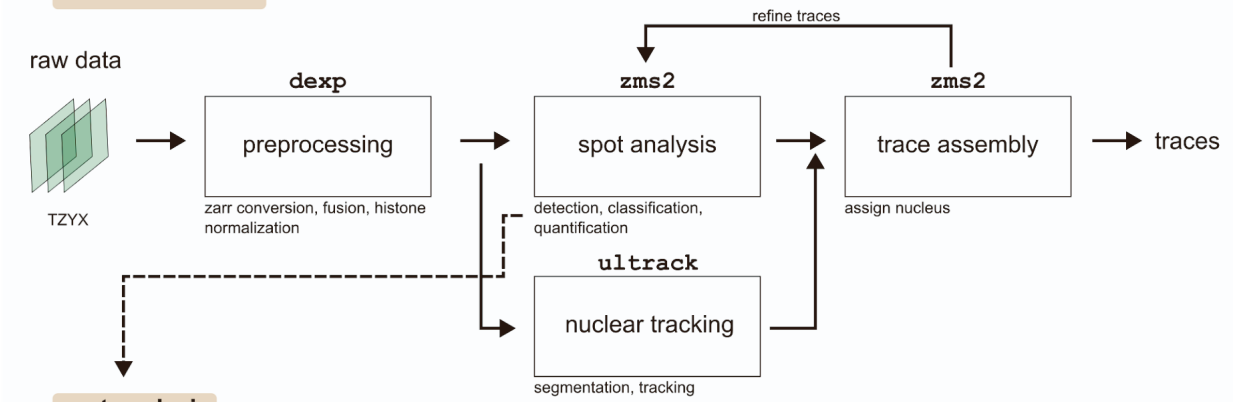
Supplemental Figure 3: Controls show overall healthy somitogenesis of *her1*-MS2 and MCP-mNeonGreen transgenic lines. (A)-(D): In situ hybridization assays against *her1* and MS2 in various genotypes at the 12-14 somite stage embryos showing qualitative agreement of the *her1* expression pattern in both wild-type and transgenic fish. All embryos are offspring of a cross of *her1*-MS2 and MCP-mNeonGreen heterozygous animals. (A) "Wild-type" embryos

lacking both markers for MS2 and MCP show normal *her1* expression patterns ⁸². (B) “*her1*-MS2” embryos carrying only the MS2 transgene show healthy *her1* expression. (C) “MCP-mNeonGreen” embryos carrying only the MCP transgene show healthy *her1* expression. (D) Embryos carrying both transgenes show healthy *her1* expression. (E) In situ hybridization assay against the MS2 sequence in *her1*-MS2 embryos showing that the reporter qualitatively recapitulates the endogenous *her1* expression pattern. (F) *In situ* hybridization assay for the somite boundary marker *xirp2a* ⁸³, showing examples of no defect, small defect, strong defect. When defects do occur, they are limited to the trunk region, which completes somite formation before our imaging experiments begin and is thus not included in our analysis. (G) The majority of animals show no defects at 32 hours post fertilization.

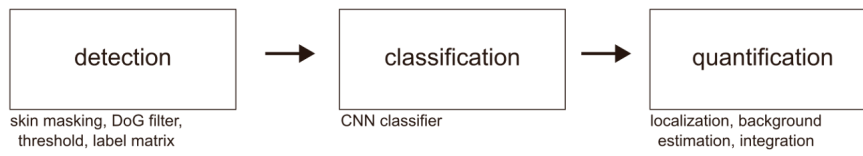


Supplemental Figure 4: Characterization of spots on commercial light sheet and confocal fluorescence microscopes. (A,B) Spots (white arrows) are detected on the Zeiss Z.1 Lightsheet but only when the reporter gene is present. MCP-mNeonGreen fish show no puncta in the absence of the *her1*-MS2 reporter. Single z-slices of embryos with MCP-mNeonGreen (green) and h2B-mScarlet (magenta) as a nuclear marker (A) with *her1*-MS2 or (B) without *her1*-MS2. Contrast was adjusted manually such that background MCP levels were approximately equal in both images. (C)-(D) Two examples of spots on a Zeiss LSM 880 confocal microscope with AiryScan processing.

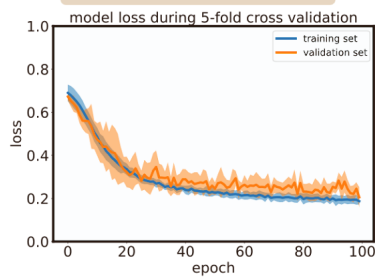
A pipeline overview



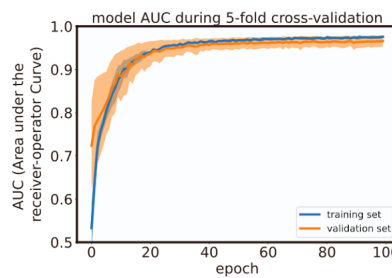
B spot analysis



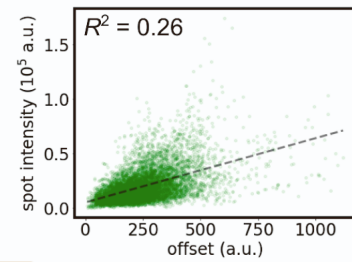
C spot classifier training



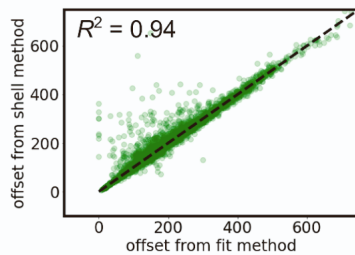
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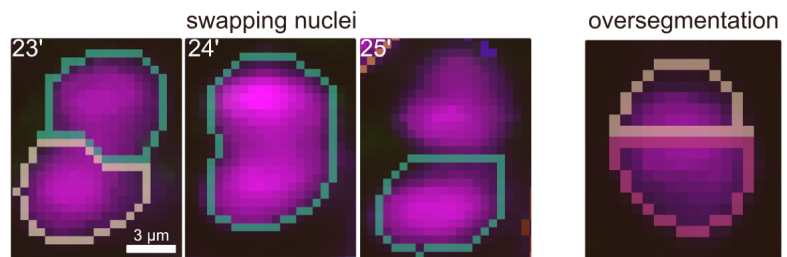
E spot intensity vs background



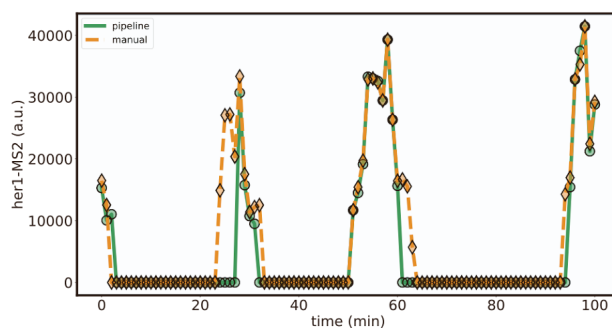
F comparison of background estimation methods



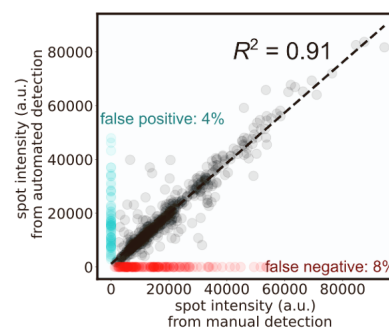
G examples of nuclear tracking errors



H trace accuracy



I

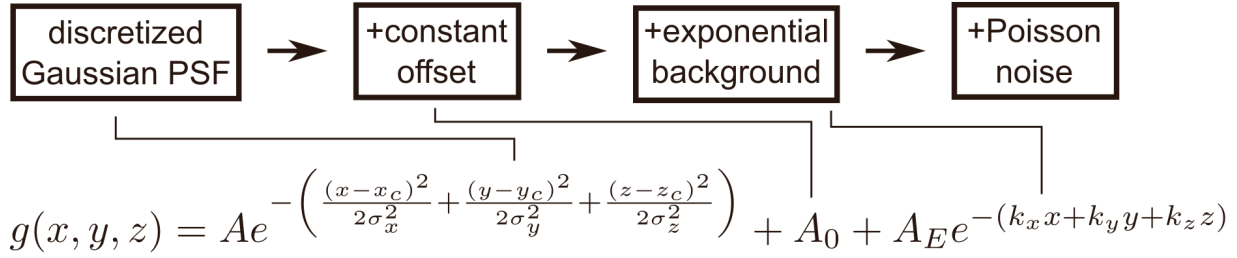


Supplemental Figure 5: Overview of image analysis pipeline and its accuracy. (A)

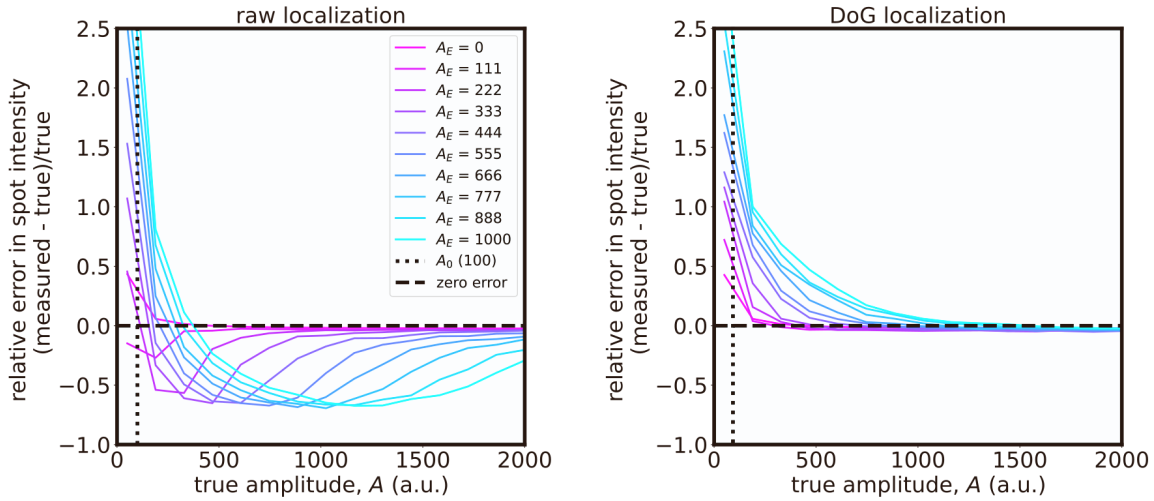
Schematic of the full pipeline. The pipeline takes two 4D image stacks as input, one for the nuclear marker and one for the MS2-MCP reporter per light sheet view. Preprocessing using the *dexp* package ⁷¹ involves conversion to the *zarr* file format, fusion of multiple light sheet views (if necessary), and normalization of the nuclear marker channel to prepare for nuclear segmentation. After preprocessing, the nuclear and spot channels are split, with the nuclear channel undergoing segmentation and tracking via *ultrack* ⁷⁴ and the spot channel passing through our spot pipeline, called *zms2*, which detects and analyzes MS2 spots. The outputs of these packages are combined into traces of spot fluorescence for a given nucleus as a function of time. Traces are then refined using an iterative computational algorithm. In brief, the algorithm identifies abrupt gaps in traces and attempts to fill them in by extracting a voxel around the expected spot location, passing it back through the classification and quantification modules, and seeing if the spot passes a set of criteria (see the “Spot analysis” section of the Methods). (B) Schematic spot analysis section of the full pipeline. Candidate spots are detected and then classified as a true spot or a false-positive detection using a binary Convolutional Neural Network (CNN) classifier. The culled list of spots then undergo quantification, where the total fluorescence intensity of the signal is computed via spot localization, background estimation and subtraction, and then summing pixel values within a defined ellipsoid (see the “Spot analysis” section of the Methods). DoG=Difference of Gaussians, CNN=Convolutional Neural Network. Skin masking=computational removal of bright and irrelevant skin cells. Thresholding occurs on the DoG-filtered image to create a binary mask, which is transformed to a label matrix that assigns a unique identifier to each candidate spot. (C) Binary cross-entropy loss function of convolutional neural network spot classifier over training shows that the model converged with minimal overfitting. Solid lines are means and shaded error bars standard deviations over 5 folds of cross validation. Blue=training set (“train”), Orange=validation set (“val”). CV=cross validation. (D) Area Under the receiver-operator Curve (AUC), a metric of model performance that balances false positive and false negative detections, during training (“epoch” = time in the training process). The validation AUC converged to 0.97 ± 0.01 , indicating very high classification performance (a perfect classifier has AUC=1, an unbiased random guess has AUC=0.5). (E) Spot intensity vs. background level (“offset”) showing an extremely weak correlation ($R^2=0.26$), which is indicative of MCP saturation and reliable measures of absolute fluorescence intensity. (F) Comparison of two methods for background estimation, described in the Methods section. “Shell” refers to taking the mean of pixel values in a shell around the spot. “Fit” refers to fitting a Gaussian with an offset. We find that the two methods give very similar results, with the data clustering closely around the 1:1 line (black dashed line) and an R^2 of 0.94. The best fit slope is 1.049 ± 0.003 . (G) Examples of nuclear tracking errors, including when the algorithm swaps the identities of nearby nuclei (left), and when it splits a single nucleus into two segments (right). Shown are single z-slices of histones (magenta), with the algorithm-assigned identity of each nucleus marked by a unique colored outline. MCP-mNeonGreen background is faintly visible in green. (H) Example comparison of a pipeline-derived trace (green solid line) and the corresponding manually-derived trace (orange dashed line), obtained by manually clicking on spots in images and passing these spots through the quantification pipeline. (I) Comparison of spot intensities between pipeline and manual traces across 728 spots from 26 nuclei. Cases

where both methods detect a spot are colored in black. Linear regression to these points shows strong correlation ($R^2 = 0.91$). Cases in which a human detected a spot and the pipeline did not (false negative) are colored in red. These cases are 8% of all spots. Cases in which a human did not detect a spot but the pipeline did (false positive) are colored in cyan. These cases are 4% of all spots.

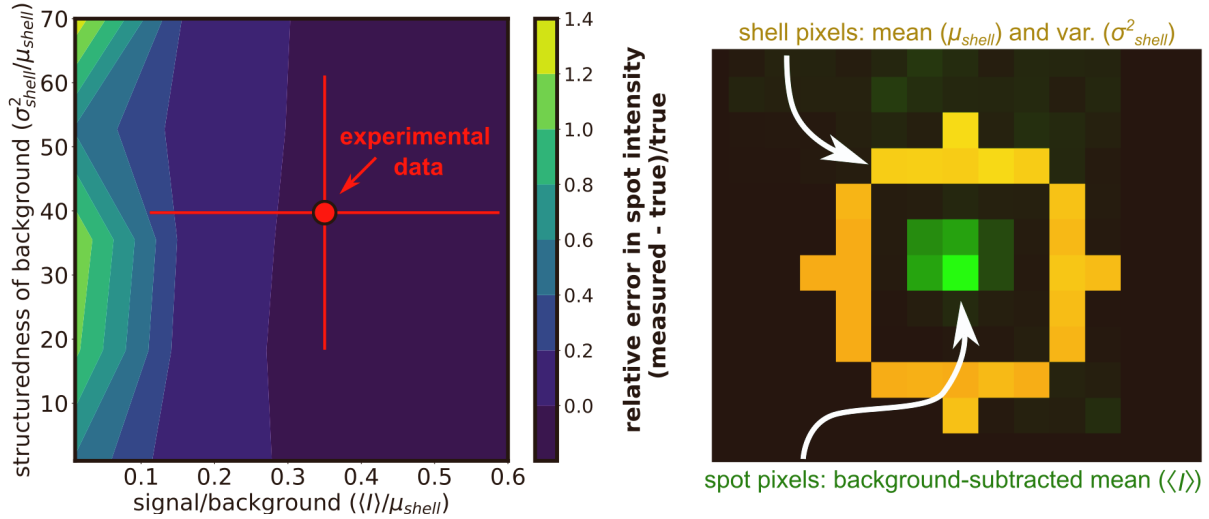
A method for generating synthetic spots



B accuracy of spot intensity measurement with raw and DoG filter-based localization

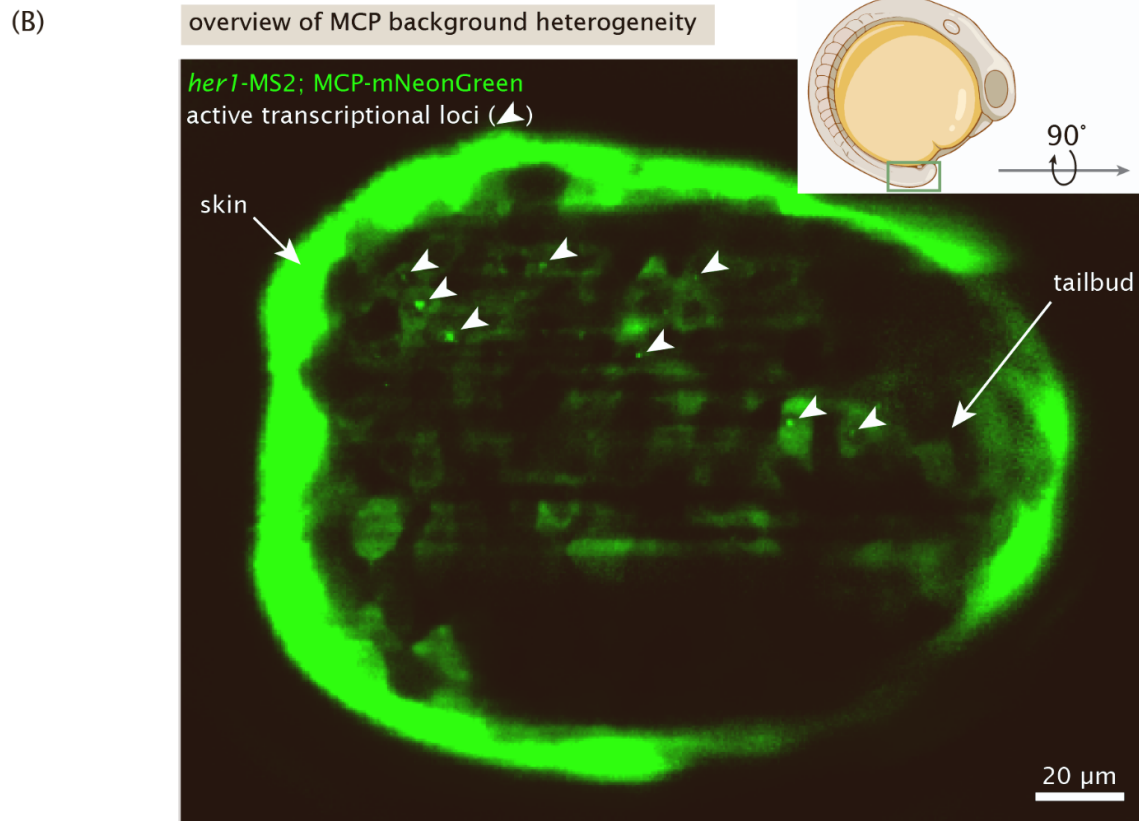
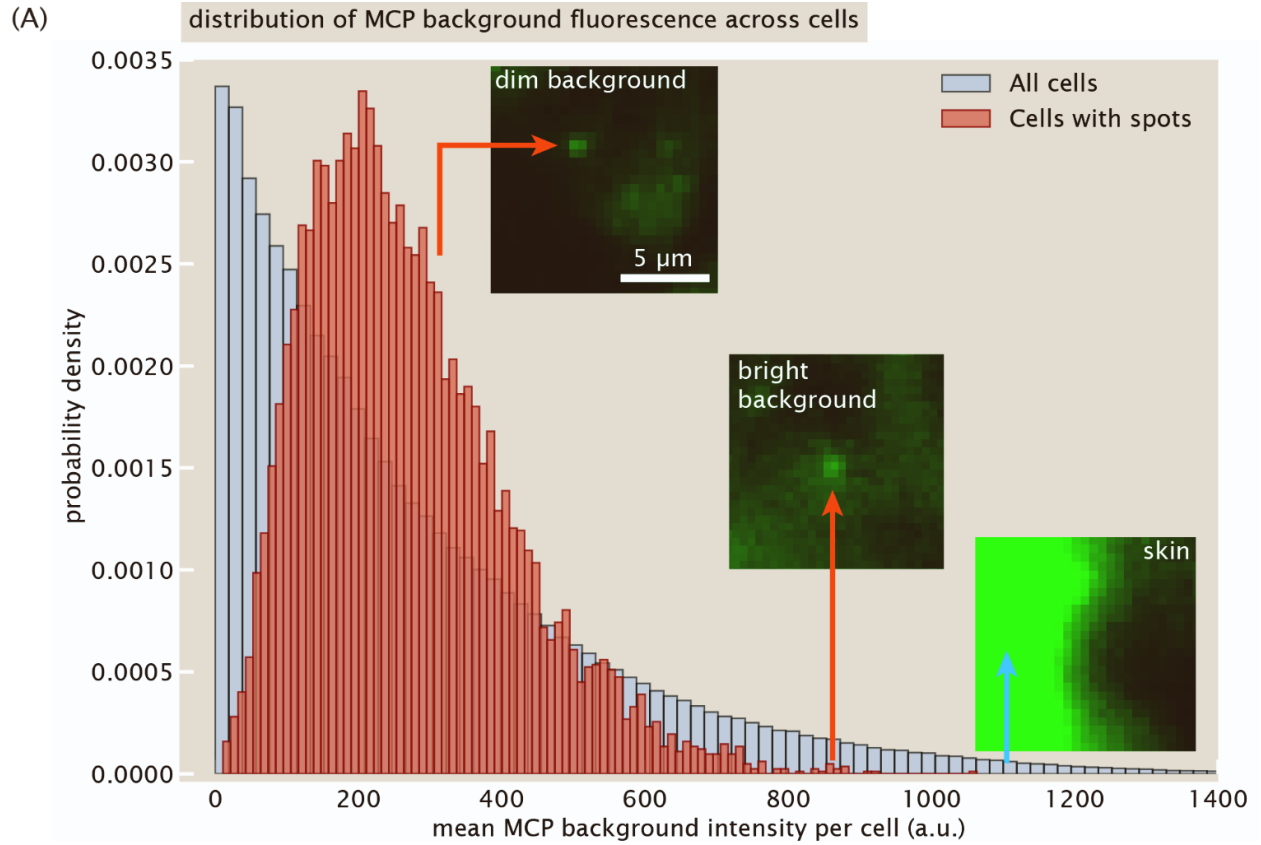


C estimated accuracy of real spots

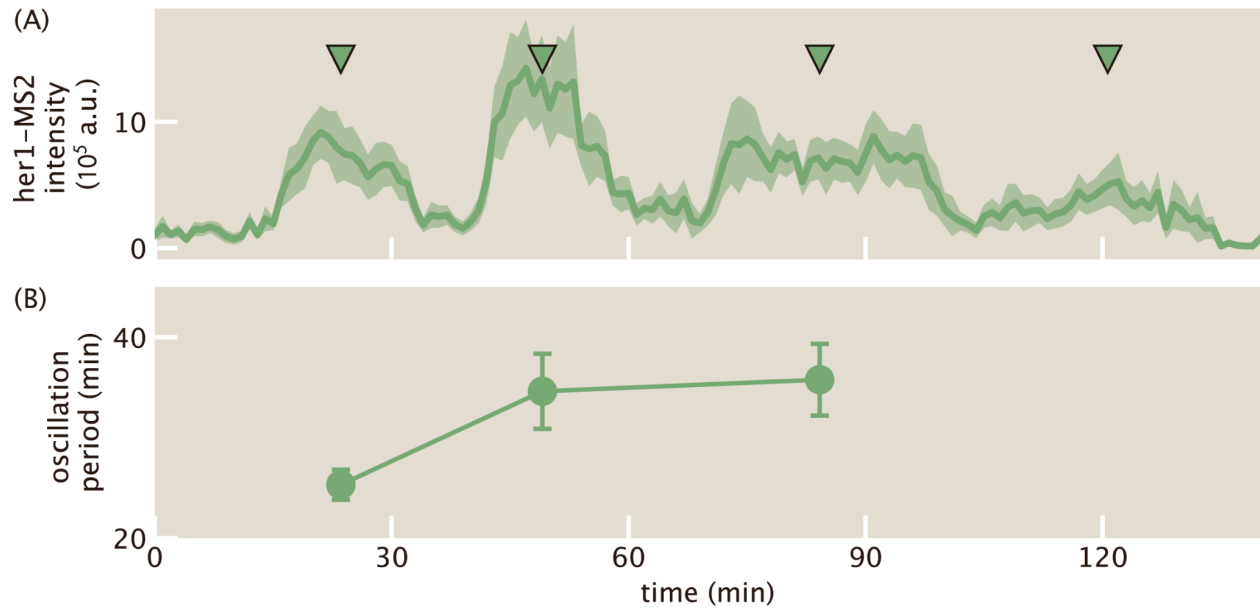


Supplemental Figure 6: Characterization of spot intensity measurement accuracy using synthetic spots. (A) Schematic of method for generating synthetic spots with structured background. The Point Spread Function (PSF) is modeled as a Gaussian function, to which we add a constant offset and an exponentially decaying background, the latter being a simple model of the structured MCP background we observe in real spots (not shown). The

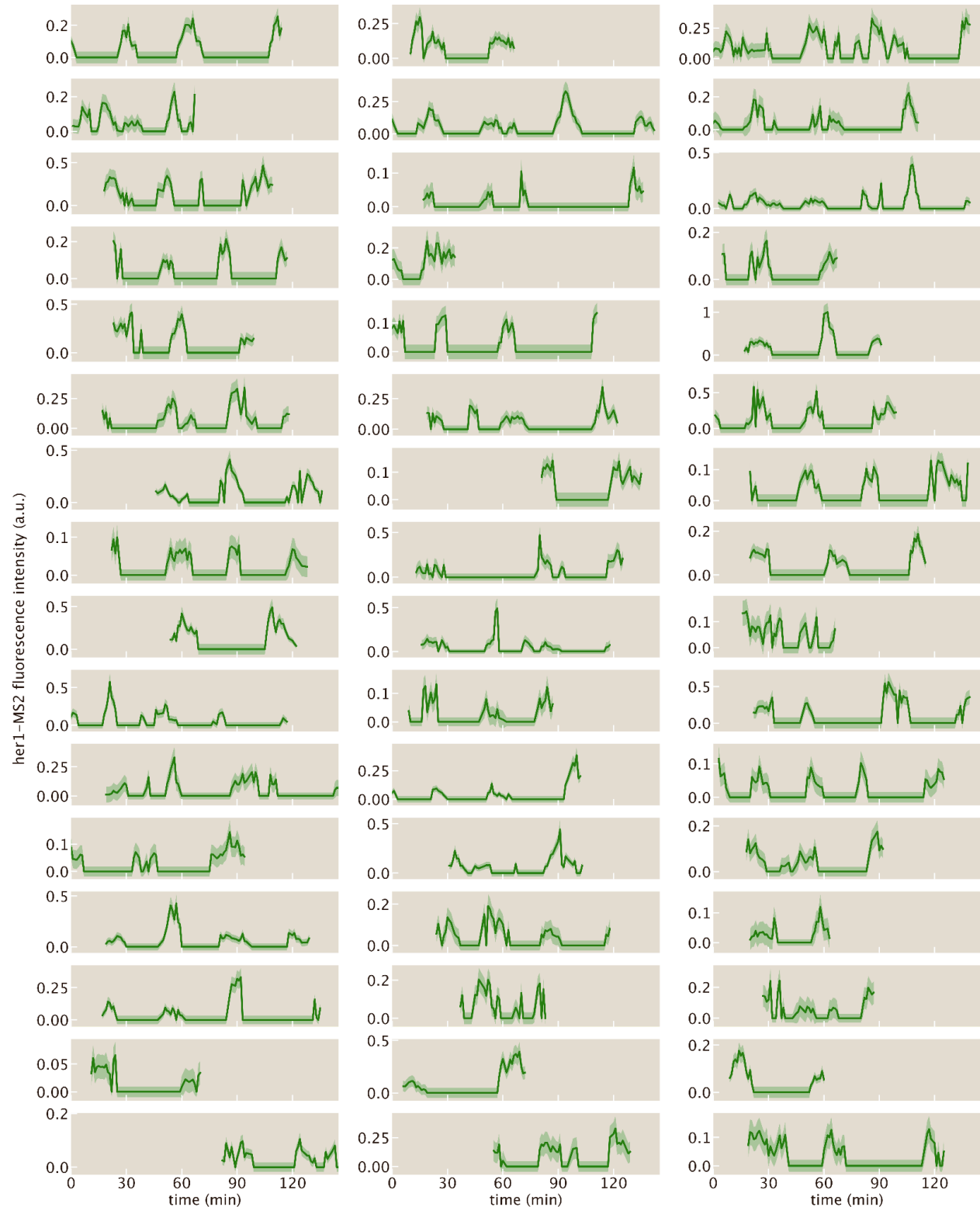
wavevector, k , of the background is taken to have a length of 10 pixels (roughly the length of the spot in voxels) and a random direction. The pixels of the final image are then drawn from a Poisson distribution with mean equal to $g(x,y,z)$. (B) Relative error in spot intensity (measured - true)/true vs true spot amplitude (the parameter A in the equation in panel (A)) for different strengths of the exponential background (via the parameter A_E). Left: spot localization via Gaussian fitting of the raw image. Right: spot localization via Gaussian fitting of a Difference-of-Gaussians (DoG)-filtered image. After localization, background estimation is done with the “shell” method described in the Methods. Intensity is then the sum of background-subtracted pixels inside a defined ellipsoid. Intensity errors are larger when localization is done on the raw pixels, due to the structured background. (C) Left: Placing the experimental data (red circle=mean, red bars=standard deviation across all spots) in a regime diagram describing relative accuracy of the intensity measurement (heatmap) as a function of signal/background and “structuredness” of the background. The x-axis, signal/background, is the average background-subtracted value of spot pixel intensities, e.g., 0.1 = 10% above background. The y-axis, structuredness of background, is the variance of shell pixel intensity divided by the mean of the shell pixel intensity. For a uniform background with Poisson noise, the structuredness will be 1. The experimental data has a background with structuredness of approximately 40 on average. The simulations from (B) were binned into a 2D histogram according to their signal/background and structuredness levels to create the final heatmap of relative error in measured spot intensity, defined as (measured intensity - true intensity)/(true intensity). For example, a relative error of 1.0 means the algorithm overestimated the total spot intensity by 100%, or a factor of 2. This heatmap illustrates that, based on our model of spots, the DoG-based localization method should lead to highly accurate intensity estimations, with a relative error of less than 10%. Right: Schematic of the parameters used to define the axes of the heatmap on the left, using a single z slice of a real spot (green) with shell pixels highlighted in orange.



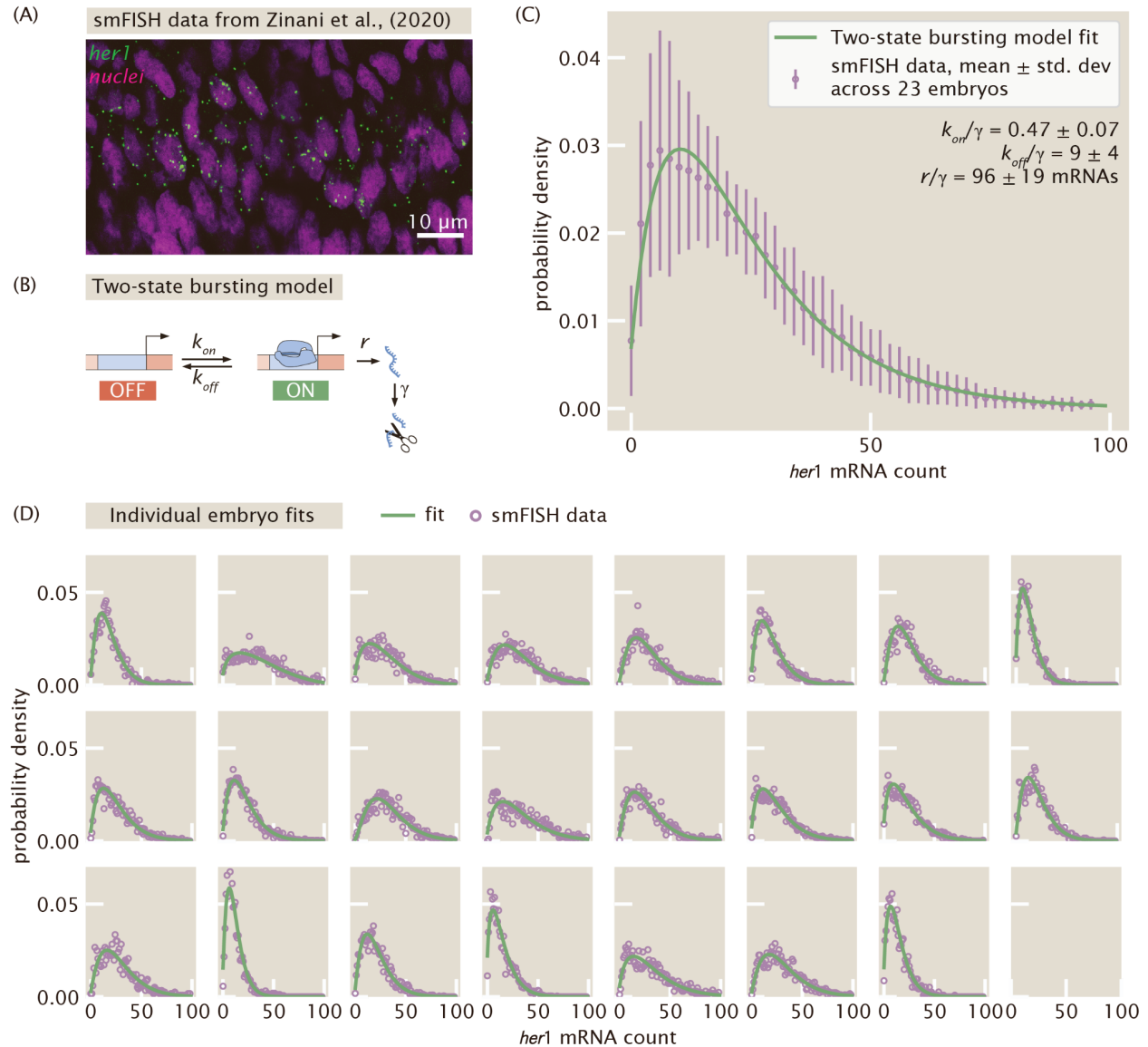
Supplemental Figure 7: MCP background levels are heterogeneous across cells, but *her1*-MS2 spots are detected across a wide range of background levels. (A) Distribution of nuclear MCP background fluorescence intensity levels across all cells (blue bars) and only cells that have active *her1*-MS2 loci (red bars). $N=31,718$ for all cells and 403 for cells with spots, which were pooled from 163 time points. *her1*-MS2 spots are detected across a wide range of background levels, excepting the very dimmest and very brightest values. The dimmest background levels may represent cells with insufficient MCP levels to generate a detectable spot, while the very brightest background levels largely correspond to skin cells, which do not express *her1*. (B) A single z-plane showing an overview of MCP heterogeneity, with active *her1*-MS2 loci marked with white arrowheads.



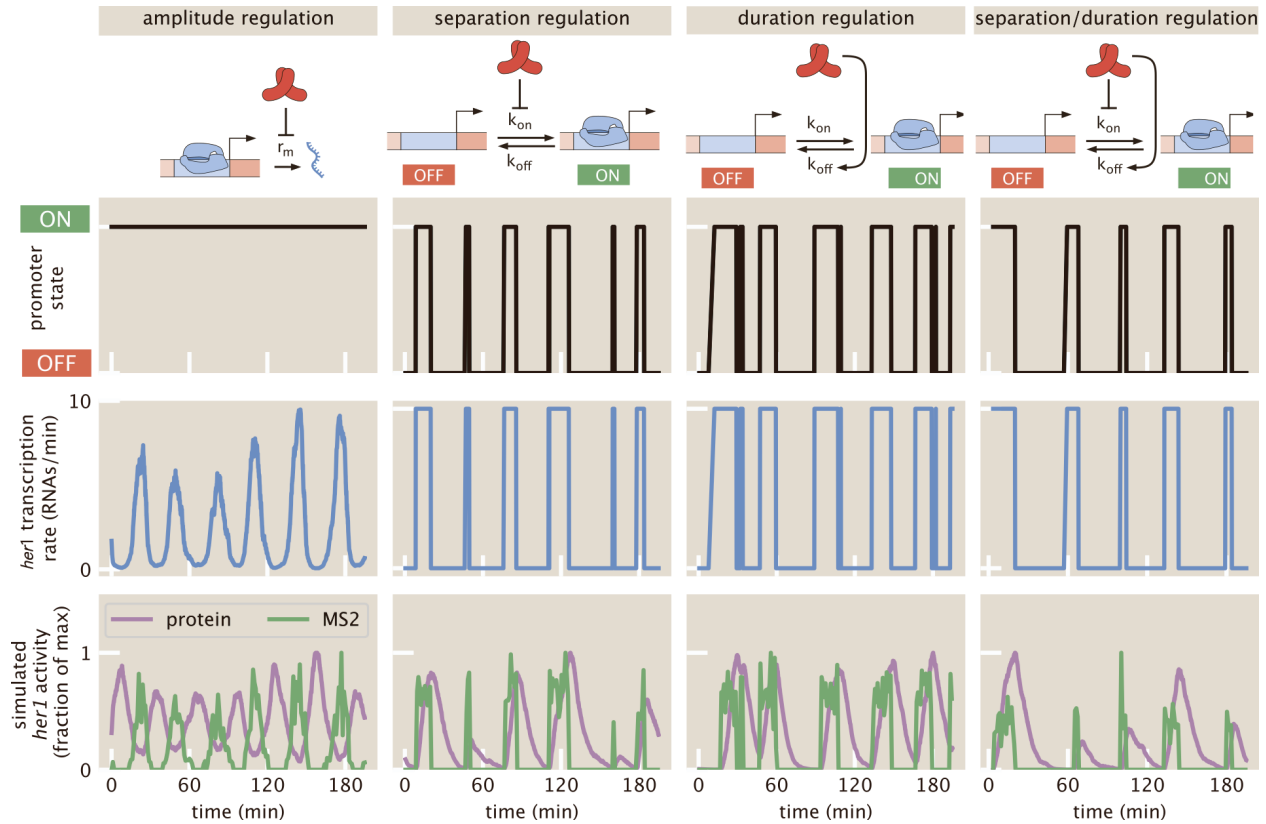
Supplemental Figure 8: Tissue-level quantification of *her1*-MS2 activity reveals oscillations with increasing period. (A) The kymograph from main text Fig. 4C was summed over the region from 240-315 μm along the anterior-posterior axis to generate a tissue-level trace that captures the dynamics prior to somite formation. The peaks of each oscillation pulse were identified by Gaussian fitting and are labelled with triangles. (B) The period between successive oscillation peaks increases as the cells approach somite formation. This is the same data that is plotted in Fig. 4E. Error bars denote standard deviations over bootstrapping.



Supplemental Figure 9. 48 randomly chosen traces with at least 20 spots per trace. Shaded error bars reflect uncertainty due to background fluctuations and spot detection (see the “Spot intensity uncertainty” section of the Methods).

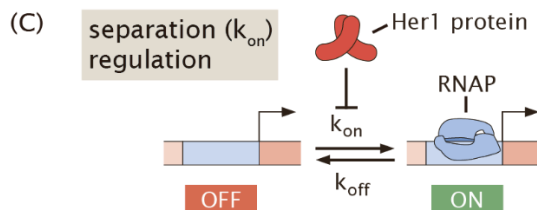
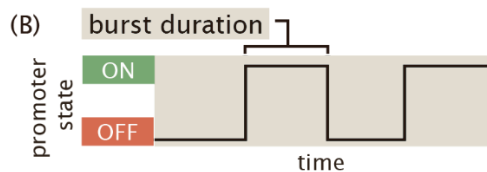
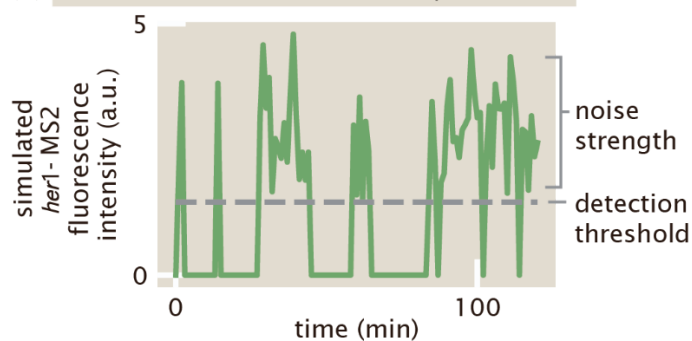


Supplemental Figure 10: A reanalysis of previously published smFISH data demonstrates that endogenous *her1* mRNA counts are consistent with a model of transcriptional bursting. (A) Example single z-plane image of smFISH data from²⁸, downloaded from the BioStudies Database with accession number S-BSST434. (B) Schematic of the canonical two-state model of transcriptional bursting (without autorepression). (C) The average probability distribution of *her1* mRNA counts across 23 embryos (magenta), overlaid with the average model fit. The mean and standard deviation of fit parameters are printed below the plot legend. (D) Data and fits for 23 individual embryos.

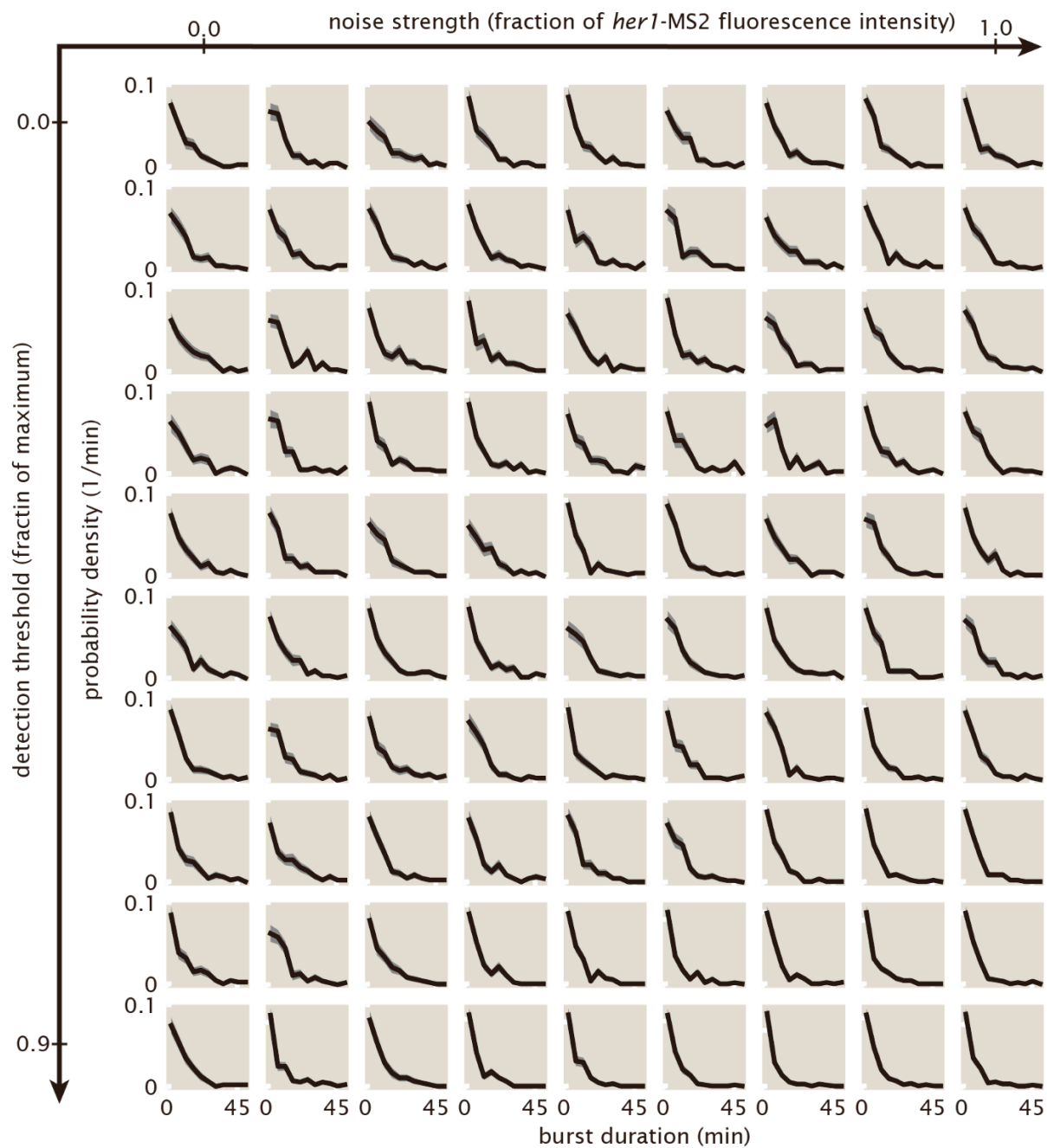


Supplemental Figure 11. Examples of simulated traces for different models. Each column corresponds to a different model of autorepression (see Fig. 6). Top row: dynamics of promoter state switching. Middle row: instantaneous transcription rate. Bottom row: simulated traces of the expected *her1*-MS2 signal (Methods) and Her1 protein concentration, normalized to maximum. Parameters are the same as in Fig. 6, described in the Methods section.

(A) simulated measurement noise parameters



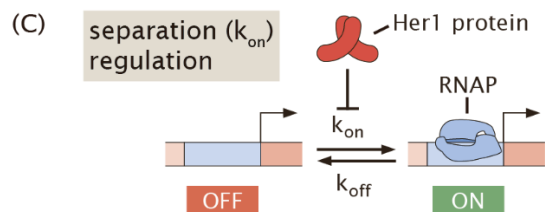
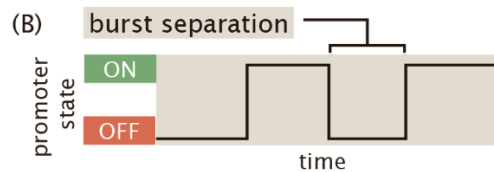
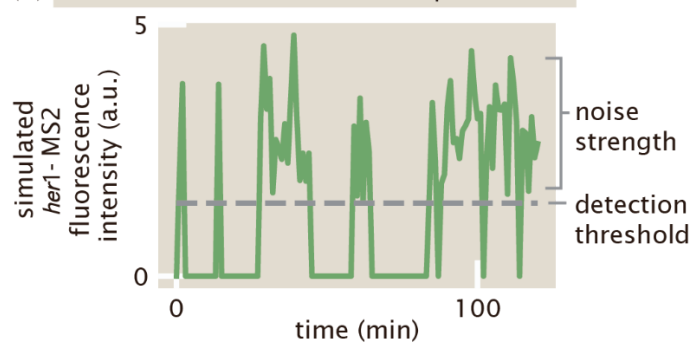
(D) burst duration distributions for separation regulation



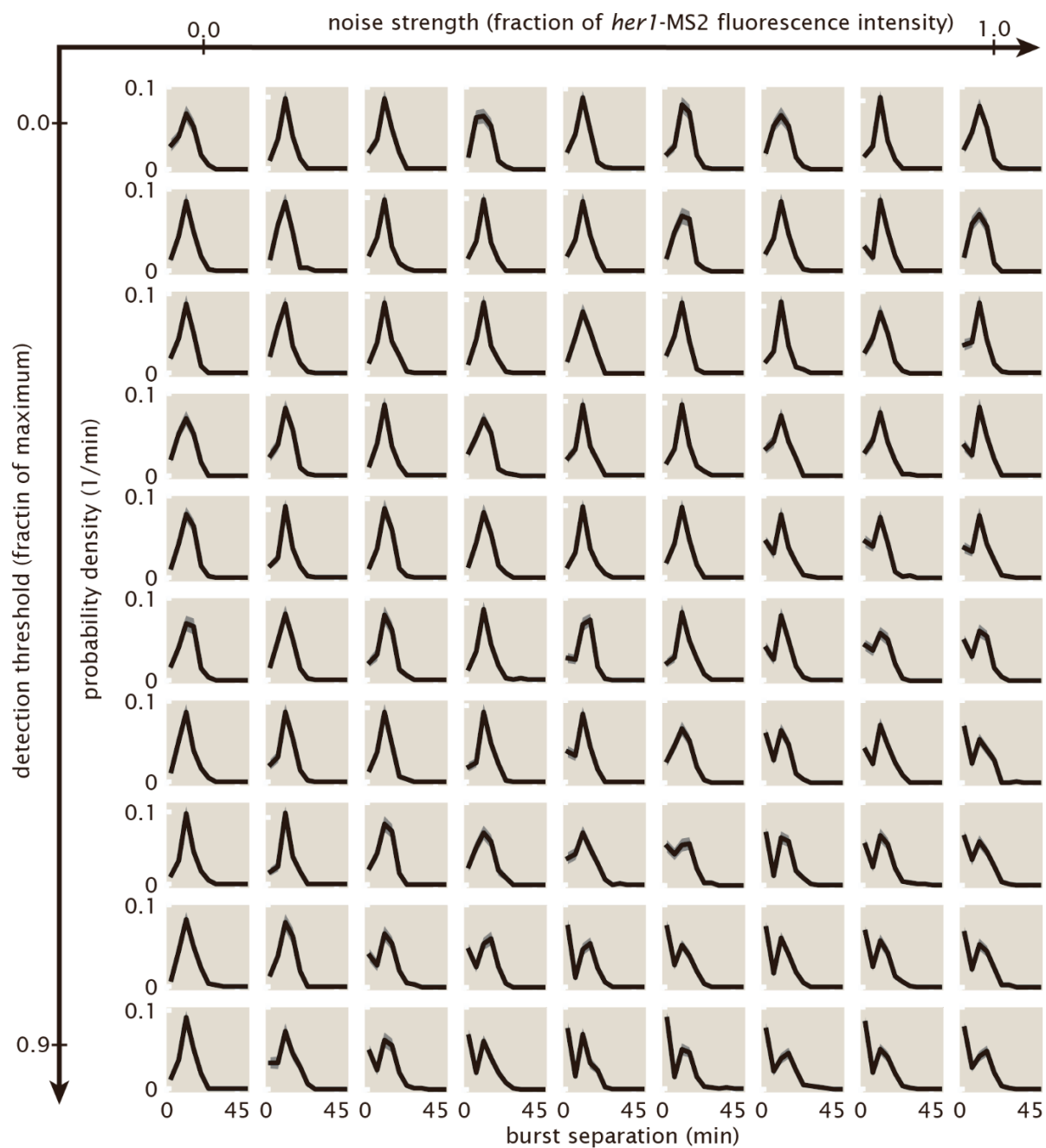
Supplemental Figure 12: Burst duration distributions for the separation regulation bursting model remain exponential for a wide range of simulated measurement noise parameters.

Our goal is to assess the impact of the MS2 measurement process on our ability to correctly identify the shape of burst interval distributions. (A) Simulated *her1*-MS2 trace illustrating the two measurement noise parameters: noise strength, which determines the level of measurement noise within a burst, and detection threshold, which determines how strong the *her1*-MS2 signal must be to be detected. Specifically, “noise strength” is the amplitude of multiplicative Gaussian noise added to the simulated *her1*-MS2 trace; we vary the noise strength from 0.0 to 1.0. The “detection threshold” is implemented by setting all values less than the detection threshold to zero; we vary the detection threshold from 0.0 to 0.9 times the maximum *her1*-MS2 signal. Other model parameters are described in the “Generalized bursting model with feedback” section of the Methods. These parameter values were chosen to produce quasi-regular oscillations while being consistent with the literature where possible. (B) In this figure, we focus on the analysis of the burst duration, defined as the duration of time the promoter is in the ON state. (C) We also focus here on the case of separation regulation, where Her1 proteins regulate the rate of the promoter turning on, k_{on} . In the absence of measurement noise, the burst duration distribution for separation regulation is exponential, since the burst duration is set by k_{off} , not k_{on} , and here k_{off} is constant. (D) We explored how measurement artifacts affect the shape of this distribution by creating a 2D grid of burst duration distributions, with increasing noise strength to the right and increasing detection threshold going down. The burst duration distribution remains exponential over all of the explored parameter space, indicating that the degree of measurement noise explored here does not change the distribution’s qualitative shape. Shaded error bars denote standard deviations over bootstrapped distributions.

(A) simulated measurement noise parameters



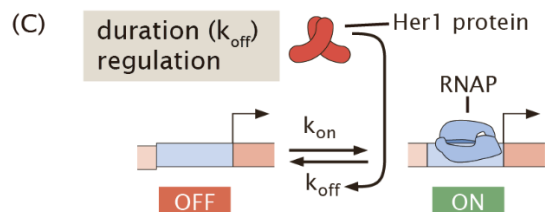
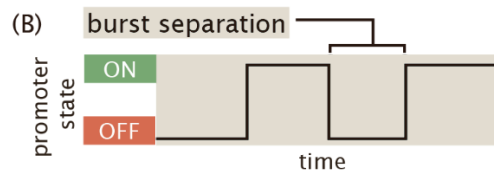
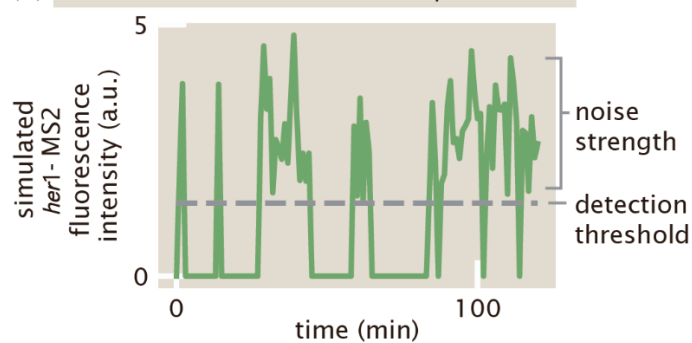
(D) burst separation distributions for separation regulation



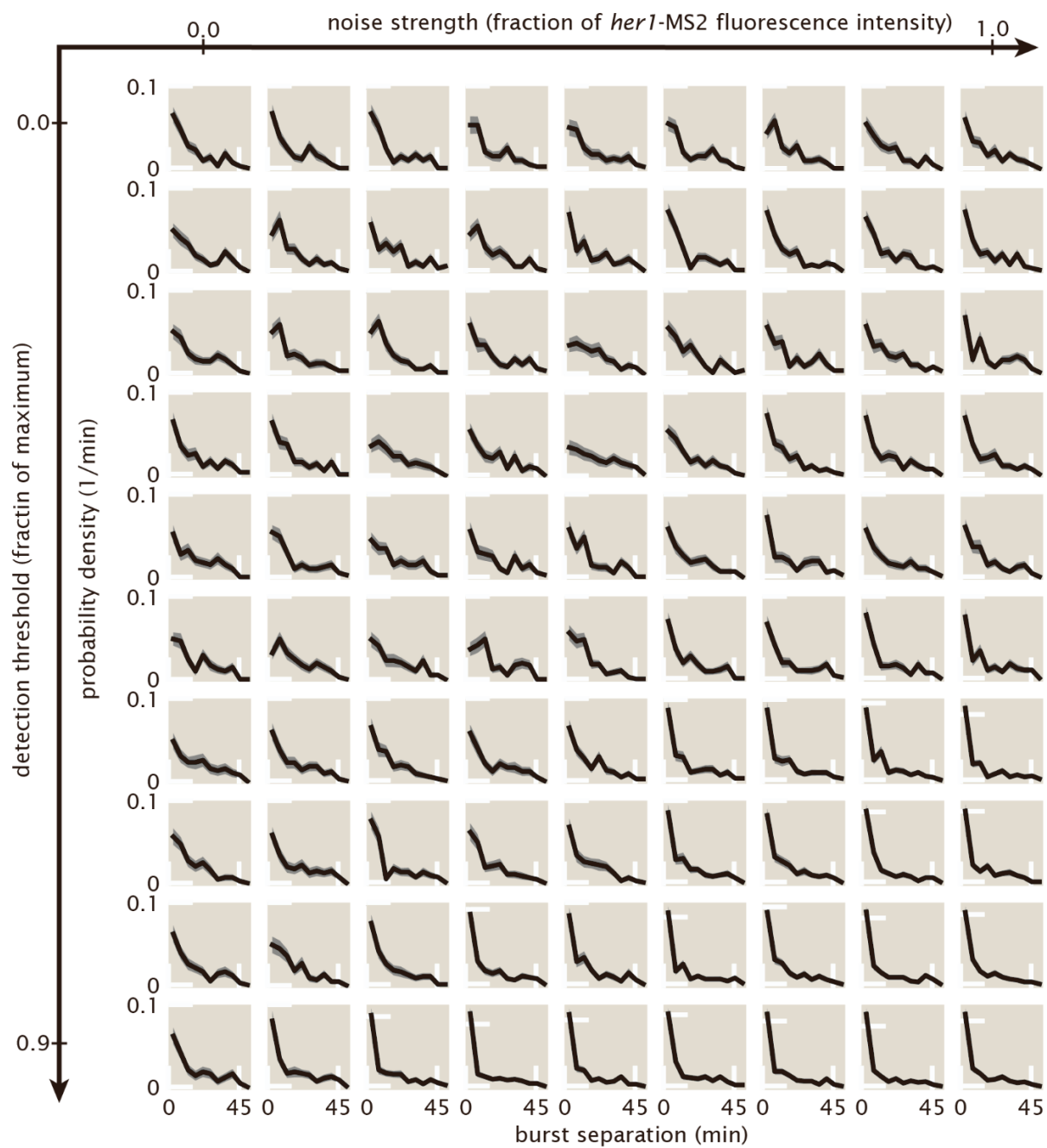
Supplemental Figure 13: Burst separation distributions for the separation regulation bursting model remain peaked for a wide range of simulated measurement noise parameters.

Our goal is to assess the impact of the MS2 measurement process on our ability to correctly identify the shape of burst interval distributions. (A) Simulated *her1*-MS2 trace illustrating the two measurement noise parameters: noise strength, which determines the level of measurement noise within a burst, and detection threshold, which determines how strong the *her1*-MS2 signal must be to be detected. Specifically, “noise strength” is the amplitude of multiplicative Gaussian noise added to the simulated *her1*-MS2 trace; we vary the noise strength from 0.0 to 1.0. The “detection threshold” is implemented by setting all values less than the detection threshold to zero; we vary the detection threshold from 0.0 to 0.9 times the maximum *her1*-MS2 signal. Other model parameters are described in the “Generalized bursting model with feedback” section of the Methods. These parameter values were chosen to produce quasi-regular oscillations while being consistent with the literature where possible. (B) In this figure, we focus on the analysis of the burst separation, defined as the duration of time the promoter is in the OFF state. (C) We also focus here on the case of separation regulation, where Her1 proteins regulate the rate of the promoter turning off, k_{off} . With our chosen parameters and in the absence of measurement noise, the burst separation distribution is peaked, reflecting the regulation of k_{on} that reduces stochasticity in burst initiation. (D) We explored how measurement artifacts affect the shape of this distribution by creating a 2D grid of burst separation distributions, with increasing noise strength to the right and increasing detection threshold going down. The distribution remains peaked over a wide range of the explored parameter space, with high noise strength and/or high detection threshold adding an exponential background to the distribution. Shaded error bars denote standard deviations over bootstrapped distributions.

(A) simulated measurement noise parameters

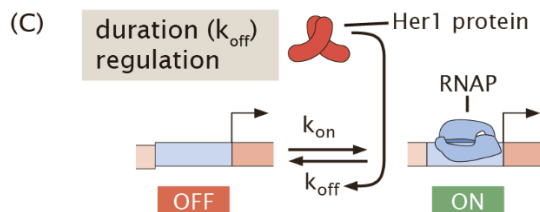
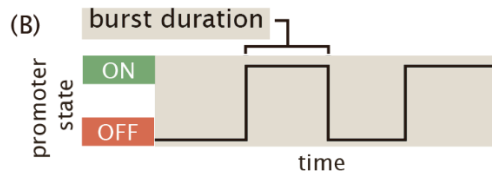
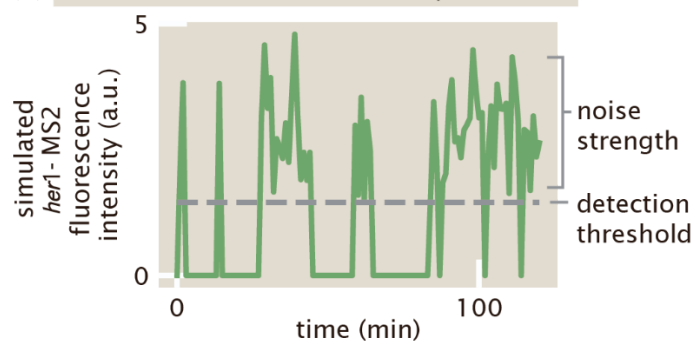


(D) burst separation distributions for duration regulation

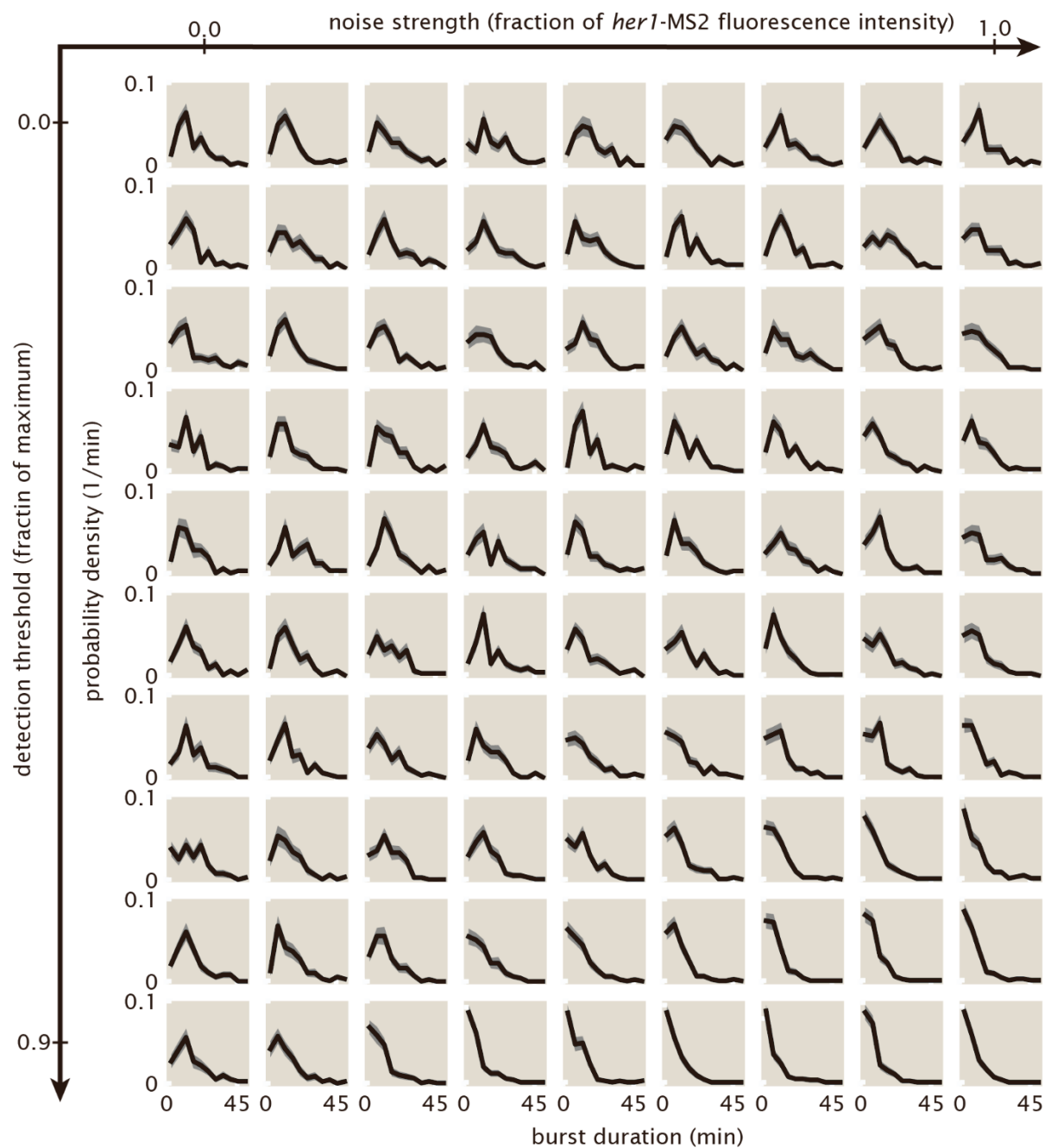


Supplemental Figure 14: Burst separation distributions for the duration regulation bursting model remain exponential for a wide range of simulated measurement noise parameters. Our goal is to assess the impact of the MS2 measurement process on our ability to correctly identify the shape of burst interval distributions. (A) Simulated *her1*-MS2 trace illustrating the two measurement noise parameters: noise strength, which determines the level of measurement noise within a burst, and detection threshold, which determines how strong the *her1*-MS2 signal must be to be detected. Specifically, “noise strength” is the amplitude of multiplicative Gaussian noise added to the simulated *her1*-MS2 trace; we vary the noise strength from 0.0 to 1.0. The “detection threshold” is implemented by setting all values less than the detection threshold to zero; we vary the detection threshold from 0.0 to 0.9 times the maximum *her1*-MS2 signal. Other model parameters are described in the “Generalized bursting model with feedback” section of the Methods. These parameter values were chosen to produce quasi-regular oscillations while being consistent with the literature where possible. (B) In this figure, we focus on the analysis of the burst separation, defined as the duration of time the promoter is in the OFF state. (C) We also focus here on the case of duration regulation, where Her1 proteins regulate the rate of the promoter turning off, k_{off} . In the absence of measurement noise, the burst separation distribution for duration regulation is exponential, since the burst separation is set by k_{on} , which is constant here. (D) We explored how measurement artifacts affect the shape of this distribution by creating a 2D grid of burst separation distributions, with increasing noise strength to the right and increasing detection threshold going down. The distribution remains exponential over all of the explored parameter space. Shaded error bars denote standard deviations over bootstrapped distributions.

(A) simulated measurement noise parameters

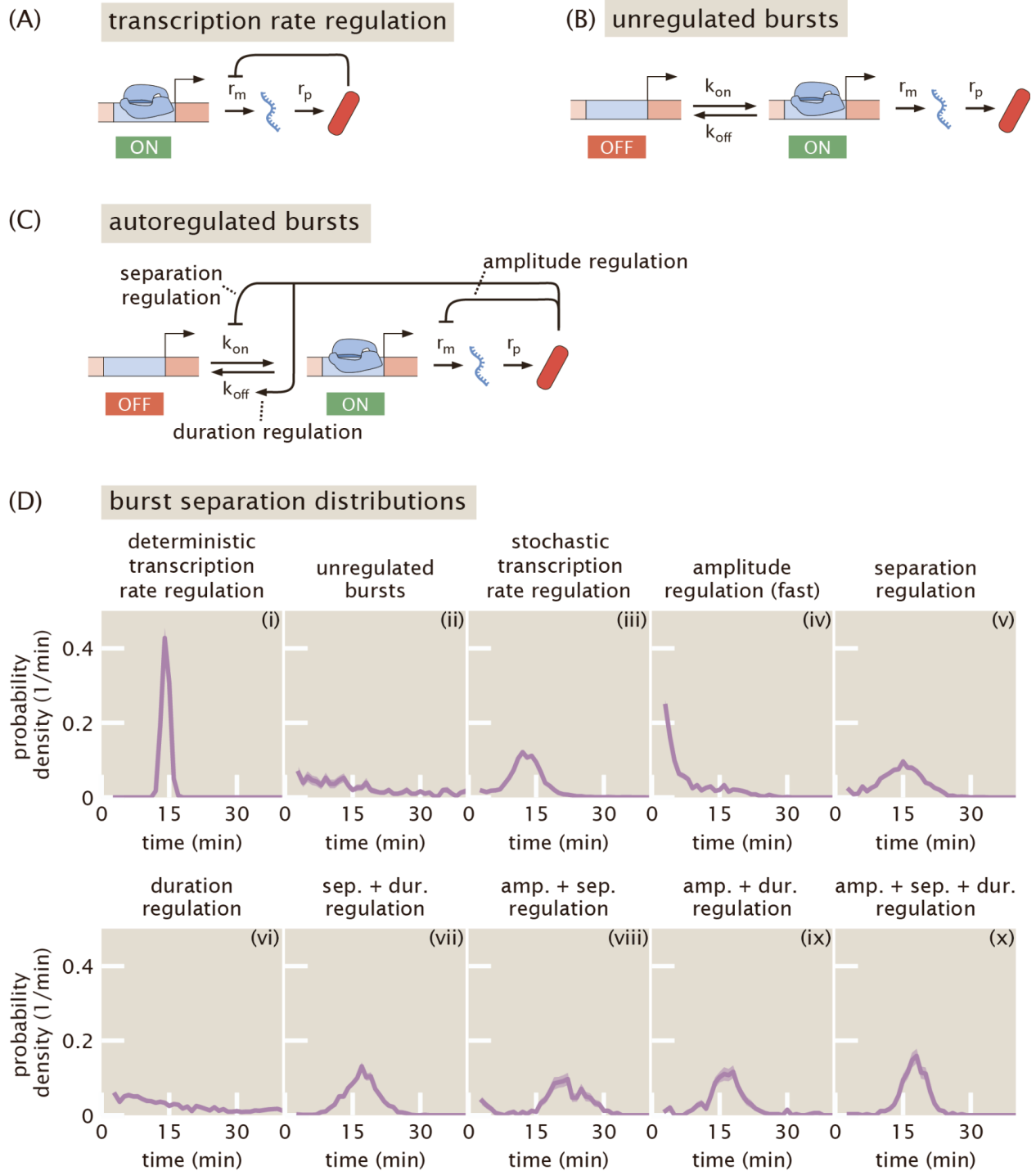


(D) burst duration distributions for duration regulation



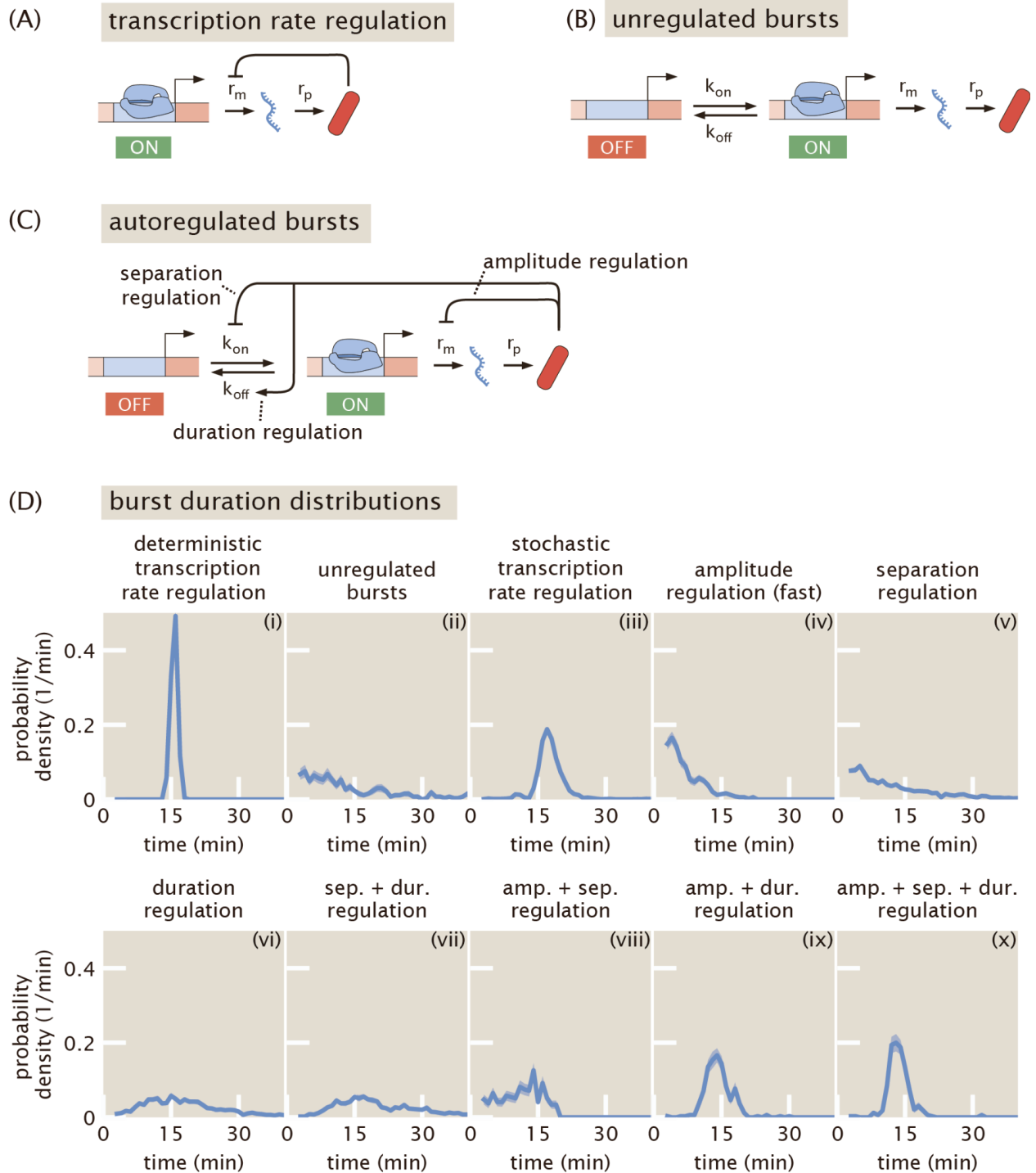
Supplemental Figure 15: Burst duration distributions for the duration regulation bursting model remain peaked for a wide range of simulated measurement noise parameters. (A)

Our goal is to assess the impact of the MS2 measurement process on our ability to correctly identify the shape of burst interval distributions. (A) Simulated *her1*-MS2 trace illustrating the two measurement noise parameters: noise strength, which determines the level of measurement noise within a burst, and detection threshold, which determines how strong the *her1*-MS2 signal must be to be detected. Specifically, “noise strength” is the amplitude of multiplicative Gaussian noise added to the simulated *her1*-MS2 trace; we vary the noise strength from 0.0 to 1.0. The “detection threshold” is implemented by setting all values less than the detection threshold to zero; we vary the detection threshold from 0.0 to 0.9 times the maximum *her1*-MS2 signal. Other model parameters are described in the “Generalized bursting model with feedback” section of the Methods. These parameter values were chosen to produce quasi-regular oscillations while being consistent with the literature where possible. (B) In this figure, we focus on the analysis of the burst duration, defined as the duration of time the promoter is in the ON state. (C) We also focus here on the case of duration regulation, where Her1 proteins regulate the rate of the promoter turning off, k_{off} . With our chosen parameters and in the absence of measurement noise, the burst duration distribution for duration regulation is peaked, reflecting the regulation of k_{off} that reduces stochasticity in burst duration. (D) We explored how measurement artifacts affect the shape of this distribution by creating a 2D grid of burst duration distributions, with increasing noise strength to the right and increasing detection threshold going down. The distribution remains peaked over a wide range of the explored parameter space, with high noise strength and/or high detection threshold adding an exponential background to the distribution that eventually dominates. Shaded error bars denote standard deviations over bootstrapped distributions.



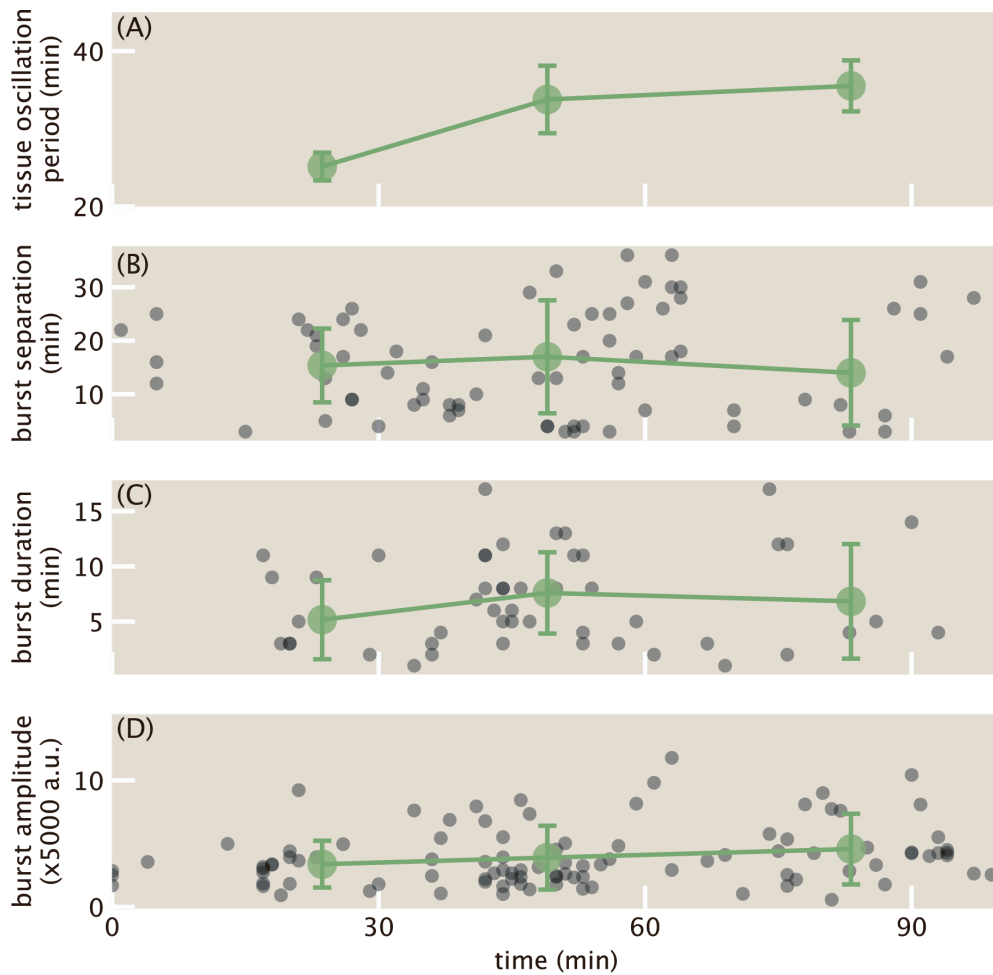
Supplemental Figure 16: Additional simulations of burst separation distributions under various regulatory mechanisms. (A)-(C) Cartoons of the various molecular control scenarios considered. (D) Burst separation distributions for various models as would be measured empirically by our *her1*-MS2 reporter. All models except unregulated bursts (ii) generate quasi-periodic solutions. Description of models: (i) deterministic transcription rate regulation—a deterministic delay differential equation where protein concentration decreases transcription rate

according to a Hill function (finite variance is due to simulated noise in the MS2 measurement; see Appendix, equations (48)-(49)); (ii) unregulated bursts—simple two-state model of transcriptional bursting without feedback; (iii) stochastic transcription rate regulation—fully stochastic version of (i); (iv)-(x) fully stochastic two-state bursting models with feedback, beginning with (iv) amplitude regulation (fast)—protein regulates burst amplitude and bursts are taken to occur on average once per minute; (v) separation regulation—protein regulates burst separation, with approximately one burst per oscillation; (vi) duration regulation—protein regulates burst duration, with approximately one burst per oscillation; (vii) separation and duration regulation—combination of (v) and (vi); (viii) amplitude and separation regulation—same as model (v) but with additional regulation of burst amplitude; (ix) amplitude and duration regulation—same as (vi) but with additional regulation of burst amplitude; (x) amplitude, separation, and duration regulation—same as (vii) but with additional regulation of burst amplitude. Parameter values for all simulations are given in the “Generalized bursting model with feedback” section of the Methods. Shaded error bars denote standard deviations over bootstrapped distributions.



Supplemental Figure 17: Additional simulations of burst duration distributions under various regulatory mechanisms. (A)-(C) Cartoons of the various molecular control scenarios considered. (D) Burst duration distributions for various models as would be measured empirically by our *her1*-MS2 reporter. All models except unregulated bursts (ii) generate quasi-periodic solutions. Description of models: (i) deterministic transcription rate regulation—a deterministic delay differential equation where protein concentration decreases transcription rate

according to a Hill function (finite variance is due to simulated noise in the MS2 measurement; see Appendix, equations (48)-(49)); (ii) unregulated bursts—simple two-state model of transcriptional bursting without feedback; (iii) stochastic transcription rate regulation—fully stochastic version of (i); (iv)-(x) fully stochastic two-state bursting models with feedback, beginning with (iv) amplitude regulation (fast)—protein regulates burst amplitude and bursts are taken to occur on average once per minute; (v) separation regulation—protein regulates burst separation, with approximately one burst per oscillation; (vi) duration regulation—protein regulates burst duration, with approximately one burst per oscillation; (vii) separation and duration regulation—combination of (v) and (vi); (viii) amplitude and separation regulation—same as model (v) but with additional regulation of burst amplitude; (ix) amplitude and duration regulation—same as (vi) but with additional regulation of burst amplitude; (x) amplitude, separation, and duration regulation—same as (vii) but with additional regulation of burst amplitude. Parameter values for all simulations are given in the “Generalized bursting model with feedback” section of the Methods. Shaded error bars denote standard deviations over bootstrapped distributions.



Supplemental Figure 18: Single-cell burst parameters fail to explain the increase in tissue-scale *her1* oscillation period. (A) Tissue-scale *her1* oscillation period, obtained by averaging the *her1*-MS2 signal over cells within spatial bins (replotted from Fig. 4E), showing a period increase between 30 and 60 min after the start of the movie, during which time cells approach somite formation. Error bars represent standard deviations over bootstrapping. (B)-(D) We look for temporal trends in burst separation (B), duration (C), and amplitude (D) that mirror the increase in period seen at the tissue level. No clear trends emerge, likely due to a limited number of samples. Each black circle represents an individual burst. Error bars in (B)-(D) denote standard deviations over single-cell bursts within each temporal bin.

Method S1: Predicting protein signal from MS2 signal

An MS2 signal is an approximation of the instantaneous mRNA production rate. Given an MS2 signal, we predict the downstream protein signal by assuming the following flow of information, from MS2 (x) to total mRNA (m) to total protein (p)

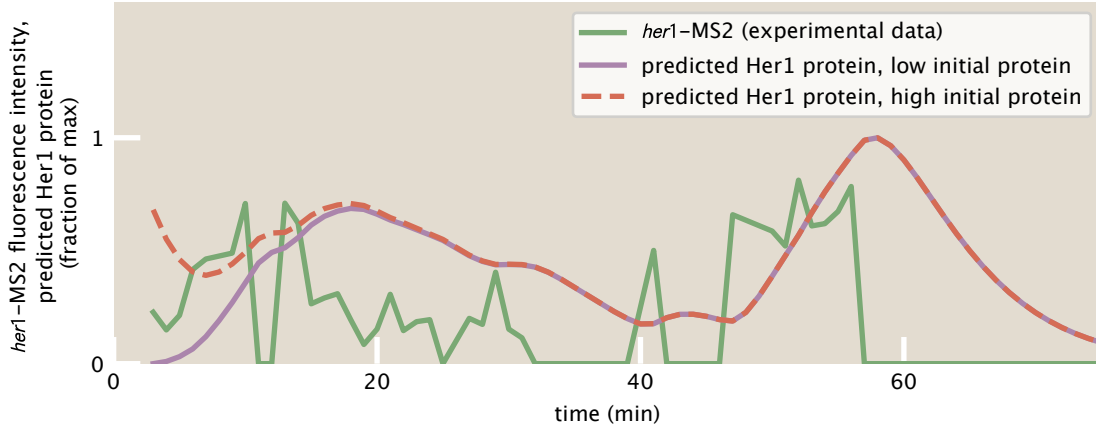
$$x = \text{given} \quad (1)$$

$$\dot{m} = r_m x - \gamma_m m \quad (2)$$

$$\dot{p} = r_p m - \gamma_p p. \quad (3)$$

Here, the r 's are production rates, and γ 's are decay rates corresponding to mRNA and protein. Given an MS2 signal and this system of equations, the predicted protein is obtained by forward integration.

One of the challenges in integrating this system of equations is that the initial conditions for mRNA and protein are unknown. Here, we show that the effect of the initial condition on the predicted protein signal diminishes exponentially with a rate approximately equal to the protein decay rate. An example of this phenomenon is shown in Supplemental Fig. 19 for a real MS2 trace. Therefore, to avoid artifacts from unknown initial conditions, we pick an arbitrary initial condition (usually zero mRNA and protein), but then throw out the predictions for mRNA and protein up to this timescale—we simply do not have the information to predict what those levels are.



Supplemental Figure 19: **Predicted protein traces for different initial conditions converge to the same solution.** In green we plot an experimental *her1*-MS2 fluorescence intensity trace. In purple is the predicted protein Her1 protein dynamics, assuming the system starts with zero initial protein. For comparison, in red is the predicted Her1 protein dynamics starting from high initial protein. After approximately one oscillation cycle, the predicted protein dynamics become insensitive to the initial condition, which is unknown in our experiment. Parameters used are $\gamma_m = 0.23 \text{ min}^{-1}$, $\gamma_p = 0.23 \text{ min}^{-1}$. Each trace was rescaled to its maximum for visual clarity.

1.1 Solving the system of equations using the Laplace transform

We show the effect of initial condition on the overall protein dynamics using the Laplace transform of the MS2 signal $x(t)$, which is given by

$$\mathcal{L}(x(t))(s) \equiv \tilde{x}(s) = \int_0^\infty dt e^{-st} x(t). \quad (4)$$

Recall the relation between Laplace transform and differentiation:

$$\mathcal{L}\left(\frac{dx}{dt}(t)\right)(s) = s\tilde{x} - x(t=0). \quad (5)$$

Using this property, we can Laplace transform the equations for m and p above, leading to

$$s\tilde{m} - m(t=0) = r_m\tilde{x} - \gamma_m\tilde{m} \quad (6)$$

$$s\tilde{p} - p(t=0) = r_p\tilde{m} - \gamma_p\tilde{p}. \quad (7)$$

Let $m(t=0) \equiv m_0$ and $p(t=0) \equiv p_0$. Solving for $\tilde{m}(s)$ and $\tilde{p}(s)$ we find

$$\tilde{m}(s) = \frac{m_0}{s + \gamma_m} + r_m \frac{\tilde{x}}{s + \gamma_m} \quad (8)$$

$$\tilde{p}(s) = \frac{p_0}{s + \gamma_p} + r_p \frac{\tilde{m}}{s + \gamma_p} \quad (9)$$

The solution for each variable has 2 pieces, one coming from the initial condition, and one coming from the signal immediately upstream: x for m , m for p . We now need to invert the Laplace transform to solve for $m(t)$ and $p(t)$. A well-known Laplace transform-inverse transform pair is

$$e^{-\gamma t} \Longleftrightarrow \frac{1}{s + \gamma}, \quad (10)$$

where the double arrow indicates the Laplace transform to the right and inverse Laplace transform to the left. With this result, the first term for $\tilde{m}(s)$ in equation 8 gives the exponential

$$\mathcal{L}^{-1}\left(\frac{m_0}{s + \gamma_m}\right) = m_0 e^{-\gamma_m t}. \quad (11)$$

The second term in equation 8 can be viewed as a product of two terms, $(s + \gamma_m)^{-1}$ and $\tilde{x}$. Like in Fourier space, in Laplace space the product of two Laplace transforms corresponds to convolution in real space. So, the inverse transform of this product is the convolution of the inverse transforms of each of the terms, resulting in

$$\mathcal{L}^{-1}\left(r_m \frac{\tilde{x}}{s + \gamma_m}\right) = r_m e^{-\gamma_m t} \star x(t). \quad (12)$$

Putting these together we have the solution

$$m(t) = m_0 e^{-\gamma_m t} + r_m e^{-\gamma_m t} \star x(t). \quad (13)$$

To be specific, the convolution term is actually an incomplete convolution, in that it only goes up to time t , namely

$$m(t) = m_0 e^{-\gamma_m t} + r_m \int_0^t dt' x(t') e^{-\gamma_m(t-t')}. \quad (14)$$

This upper limit reflects the causality of the differential equation, i.e., mRNA at time t is only determined by MS2 signal up to time t .

Analogously, the solution for $p(t)$ is

$$p(t) = p_0 e^{-\gamma_p t} + r_p \int_0^t dt' m(t') e^{-\gamma_p(t-t')}. \quad (15)$$

Plugging in the solution for $m(t)$, we get

$$p(t) = p_0 e^{-\gamma_p t} + m_0 r_p \frac{1}{\gamma_m - \gamma_p} (e^{-\gamma_p t} - e^{-\gamma_m t}) + r_m r_p \int_0^t dt' \int_0^{t'} dt'' x(t'') e^{-\gamma_m(t'-t'')} e^{-\gamma_p(t-t')}. \quad (16)$$

The third term contains convolutions with decaying exponentials of rates given by the molecular decay rate γ_m and γ_p . This part reflects the low-pass filtering done by the process of accumulating and destroying mRNA and protein. The first two terms reflect the initial condition, and we see explicitly that these terms decay to zero exponentially on the timescale set by the decay rates.

For the segmentation clock, the mRNA and protein decay rates are approximately equal. Setting them equal here, we simplify to

$$p(t) = p_0 e^{-\gamma_p t} + m_0 r_p t e^{-\gamma_p t} + r_m r_p \int_0^t dt' \int_0^{t'} dt'' x(t'') e^{-\gamma_p(t-t'')}. \quad (17)$$

Thus, as $t \gg 1/\gamma_p$, the solution converges to a function that is independent of the initial condition, as shown in Supplemental Fig. 19.

Method S2: MS2 trace uncertainty

2.1 Uncertainty due to background fluctuations

We quantify spot intensity by finding the spot center, estimating the background intensity levels around the spot, subtracting this background from each spot pixel (enforcing non-negativity), and then summing the pixels contained in an area defined by the spot size (i.e., by the point spread function).

Mathematically, the total spot intensity, I , is given by a sum of individual pixel intensities, I_j minus the background, B :

$$I = \sum_{j \in \mathcal{V}} (I_j - B), \quad (18)$$

where the sum is over pixels contained in a neighborhood volume $\mathcal{V}$ around the spot center. Let there be N pixels in $\mathcal{V}$. Then we can write I as

$$I = \left(\sum_{j=1}^N I_j \right) - NB. \quad (19)$$

If we assume that the fluctuations in each pixel level, I_j , are all independent from one another (as is the typical model of shot noise in a microscope), and are independent from the background level, B , then the uncertainty in our measurement of the total spot intensity, I , is given by,

$$\sigma_I^2 = \sum_{j=1}^N \sigma_{I_j}^2 + N^2 \sigma_B^2. \quad (20)$$

The assumption that is typically made [1] is that the σ_{I_j} s are all the same and are equal to σ_B : that there is one typical scale of fluorescence fluctuations that affect the background and signal identically. Given this assumption, we then estimate σ_B by looking at the fluctuations in the background. This is done by fitting a smooth trend to the time trace of the background signal (where the background level is obtained, for instance, by Gaussian fitting). A representative example of this background over time, together with its smooth polynomial fit, is shown in Supplemental Fig. 20A. Then, the RMS deviation of the background from this smooth trend is measured, resulting in the error bars shown in Supplemental Fig. 20B for non-zero *her1*-MS2 spot fluorescence intensities; the error bars on points with a value of zero come from spot detection uncertainties, as described below). Through 5-fold cross-validation, we found that fitting a polynomial of degree 4 to the background trend gives the lowest root-mean-squared-error fit without over-fitting (Supplemental Fig. 20C).

With this estimate of σ_B , we get

$$\sigma_I^2 = \sum_{j=1}^N \sigma_B^2 + N^2 \sigma_B^2 = \sigma_B^2 (N + N^2). \quad (21)$$

Rearranging, we arrive at

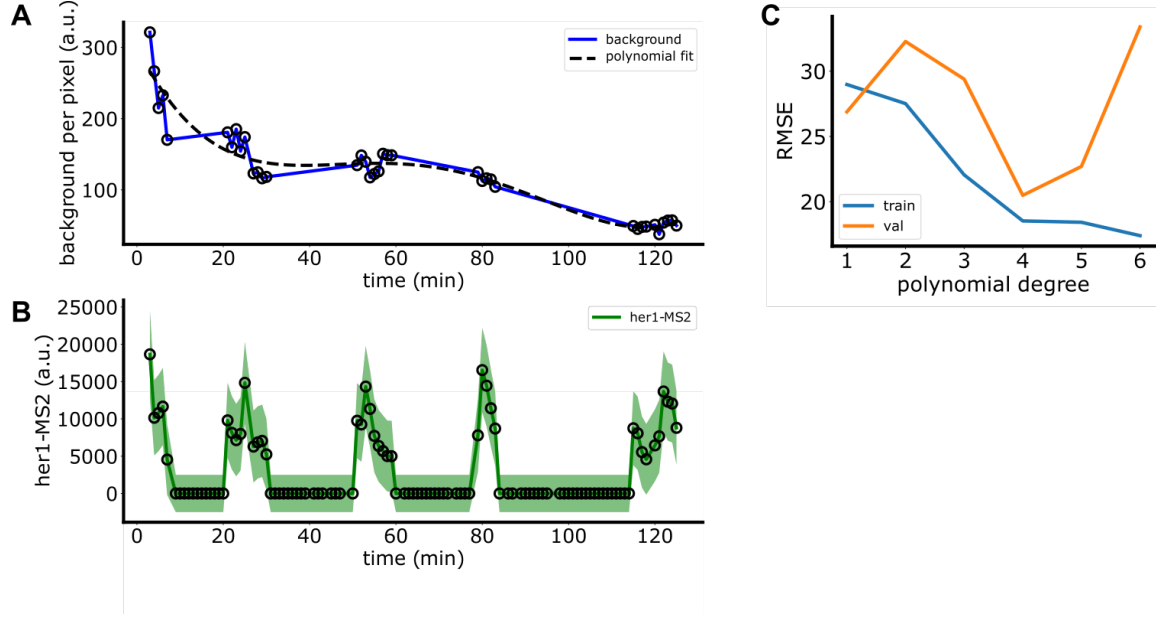
$$\sigma_I = \sigma_B N \left(1 + \frac{1}{N} \right)^{1/2}. \quad (22)$$

We use this equation to assign uncertainty to each spot fluorescence measurement.

2.2 Including detection uncertainty

If we are relying on automated spot detection, there is also some additional uncertainty due to the probabilities of false positive and false negative detections. Here we incorporate those sources of uncertainty. The approach is to define a compound stochastic process that combines the probability of false positive/false negative detections with the measurement uncertainty obtained above.

Let the false positive probability be f_p and the false negative probability be f_n . These are false positive and false negative rates for a whole dataset, measured by comparing a subset



Supplemental Figure 20: **Example of background fluctuation measurement.** (A) The mean background per pixel plotted over time for a given trace. Dashed line shows a polynomial fit of degree 4. (B) *her1*-MS2 trace with error bars derived from the background fluctuations as well as spot detection. (C) Root-mean-squared-error (RMSE) of the polynomial fit from (A) in a 5-fold cross-validation scheme, showing the train (blue) and validation (orange) errors. The minimum validation error occurs at polynomial degree 4.

of traces to manually obtained traces, i.e. ground truth for the presence/absence of a spot. We break up the analysis into the case where we are adding uncertainty to detected spots and the case where we are adding uncertainty to a “zero” (no spot).

Case 1: a spot is detected

Suppose we detect a spot and measure its intensity I and uncertainty σ_I . There is a probability f_p that the spot is a false detection, and that the true intensity of the trace at this time point is zero. We want to incorporate this uncertainty.

Define a binary variable

$$\eta = \begin{cases} 1 & \text{if the spot is real (probability } 1 - f_p) \\ 0 & \text{if the spot is a false positive (probability } f_p). \end{cases} \quad (23)$$

Now define a stochastic variable X that combines the intensity I , which we treat as a random variable, and the hidden “truth” variable η :

$$X = \begin{cases} I & \text{if } \eta = 1 \\ 0 & \text{if } \eta = 0. \end{cases} \quad (24)$$

The full uncertainty in our measurement of the intensity at this time point is the standard deviation of X . To calculate this uncertainty, we use the law of total variance, which states

that

$$\text{Var}X = \mathbb{E}[\text{Var}(X|\eta)] + \text{Var}(\mathbb{E}[X|\eta]). \quad (25)$$

The first term, the average variance, is computed as follows. If $\eta = 1$, $X = I$, and $\text{Var}X = \sigma_I^2$. If $\eta = 0$, $X = 0$, and $\text{Var}X = 0$ (the intensity of a ‘no spot’ is exactly 0). The average variance is then given by

$$\begin{aligned} \mathbb{E}[\text{Var}(X|\eta)] &= \text{Var}(X|\eta = 1)P(\eta = 1) + \text{Var}(X|\eta = 0)P(\eta = 0) \\ &= \sigma_I^2(1 - f_p) + 0 * f_p = \sigma_I^2(1 - f_p), \end{aligned} \quad (26)$$

where in the first line $P(\eta = 1)$ is the probability that $\eta = 1$, which is $1 - f_p$, and analogously for the $\eta = 0$.

The second term, the variance of the averages, is similarly reasoned: if $\eta = 1$, $X = I$, and given that we only have one measurement of I , we assume $\mathbb{E}[X] = I$. If $\eta = 0$, $X = 0$, and $\mathbb{E}[X] = 0$. For convenience, define $M := \mathbb{E}[X|\eta]$. The average of M is given by

$$\bar{M} = I(1 - f_p) + 0 * f_p = I(1 - f_p). \quad (27)$$

Now we can compute

$$\begin{aligned} \text{Var}(\mathbb{E}[X|\eta]) &= \mathbb{E}[(M - \bar{M})^2] \\ &= (I - I(1 - f_p))^2(1 - f_p) + (0 - I(1 - f_p))^2 f_p \\ &= I^2 f_p^2 (1 - f_p) + I^2 (1 - f_p)^2 f_p \\ &= I^2 f_p (1 - f_p) (f_p + 1 - f_p) \\ &= I^2 f_p (1 - f_p). \end{aligned} \quad (28)$$

Putting these two results together, we arrive at

$$\text{Var}X = \sigma_I^2(1 - f_p) + I^2 f_p(1 - f_p), \quad (29)$$

or

$$\sigma_X = \sqrt{\sigma_I^2(1 - f_p) + I^2 f_p(1 - f_p)}. \quad (30)$$

Case 2: no spot is detected

Here, we measure a zero, but there is a probability f_n that we missed a true spot with non-zero intensity. We will estimate the ‘missing’ intensity by the empirical mean of the trace, $\bar{I}$ —an exact number, not a stochastic one.

The approach here mirrors case 1, but is simpler, because the ‘missing’ value is exact. Define

$$\eta = \begin{cases} 1 & \text{if there is really no spot (probability } 1 - f_n) \\ 0 & \text{if we missed a spot (i.e. false negative with probability } f_n). \end{cases} \quad (31)$$

In this case, $\text{Var}(X|\eta) = 0$ for both values of η , so we simply have

$$\text{Var}X = \text{Var}(\mathbb{E}[X|\eta]). \quad (32)$$

Let $M := \mathbb{E}[X|\eta]$. Its average

$$\bar{M} = \bar{I}f_n. \quad (33)$$

Then we have

$$\begin{aligned} \text{Var} &= (\bar{I} - \bar{I}f_n)^2 f_n + (\bar{I}f_n)^2 (1 - f_n) \\ &= \bar{I}^2 (1 - f_n)^2 f_n + \bar{I}^2 f_n^2 (1 - f_n) \\ &= \bar{I}^2 f_n (1 - f_n) (1 - f_n + f_n) \\ &= \bar{I}^2 f_n (1 - f_n). \end{aligned} \quad (34)$$

In terms of the final uncertainty,

$$\sigma_X = \bar{I} \sqrt{f_n (1 - f_n)}. \quad (35)$$

Conclusion

Putting all these results together, we arrive at the final uncertainty of our measurement

$$\sigma = \begin{cases} \sqrt{\sigma_I^2 (1 - f_p) + I^2 f_p (1 - f_p)} & I > 0 \\ \bar{I} \sqrt{f_n (1 - f_n)} & I = 0 \end{cases} \quad (36)$$

where σ_I is the uncertainty from background fluctuations, f_p is the false-positive rate of detection, f_n is the false-negative rate of detection, I is the intensity of the individual spot, and $\bar{I}$ is the mean intensity of the trace.

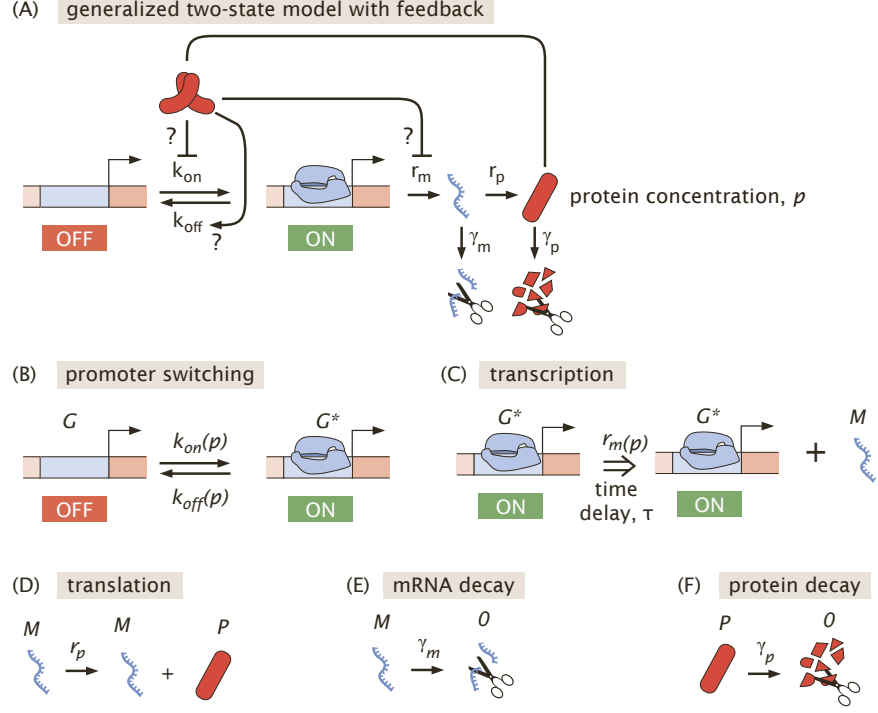
Method S3: Generalized transcriptional bursting model with feedback

In this section we provide details on the mathematical model from Fig. 6 of the main text. The goal of the model is to provide a minimal phenomenological description of transcriptional bursting with auto-repression that is flexible enough to include a range of possible regulatory mechanisms.

3.1 Statement of the model

The model is fully stochastic and is described by the following set of chemical reactions:





Supplemental Figure 21: **Overview of the bursting model.** (A) Cartoon of the overall model structure. The *her1* gene switches between ON and OFF states. In the ON state, mRNA is produced, which is translated into protein. The Her1 protein has the ability to regulate the ON-OFF switching rates and/or the transcription rate. Both mRNA and proteins also decay. (B)-(F) Cartoons showing the individual reactions that make up the model: (B) gene state switching; (C) transcription, which is a delayed reaction that takes a time τ to complete; (D) translation; (E), mRNA decay; (F) protein decay. In these cartoons, variables in capital letters— G , G^* , M , and P —represent individual molecules, while lowercase p denotes the concentration of proteins.

The variables G and G^* represent the OFF and ON promoter states, respectively. The variables M and P schematically represent individual mRNA and protein molecules, respectively, while p denotes the number of P molecules. The rates k_{on} , k_{off} , and r_m are generically functions of p and encode the types of autoregulation that occur. The double arrow notation $\Rightarrow_\tau$ denotes a delayed reaction that takes a time τ to complete. Computationally, the delayed transcription reaction is initiated with a rate that depends on the current protein concentration, but the new mRNA is not added to the system until a time τ later, following [2]. This time delay is crucial for generating oscillations under the amplitude regulation scheme, but is not necessary for frequency or duration regulation schemes. Thus, the general model is a delayed stochastic system that is simulated using the delayed Gillespie algorithm of Bratsun et al. [2] implemented in Python.

Amplitude regulation:

We model amplitude regulation via a Hill function

$$r_m(p) = r_{m,0} + r_{m,1} \frac{1}{1 + (p/K_D)^n}. \quad (42)$$

The transcription rate has a basal rate, $r_{m,0}$, and a protein-dependent rate with prefactor $r_{m,1}$. The protein-dependence is described by a decreasing Hill function with coefficient, n , and scale K_D . For simplicity, we set $r_{m,0} = 0$. Obtaining robust oscillations under the amplitude regulation model requires addition time delay to be introduced into the system. We do this by explicitly adding a time delay, τ , between the initiation of the mRNA production reaction and its completion within the simulation [2]. In this model, the rate of transcription initiation depends on the current number of Her1 proteins, but the time required to produce mature *her1* mRNAs is modeled explicitly.

Under amplitude regulation, switching between G and G^* is not required to generate oscillations. Rather, the sharp modulation of the transcription rate by Her1 proteins on their own can produce burst-like behavior.

Frequency regulation:

We model frequency regulation via

$$k_{on} = k_{+,0} + k_{+,1} \frac{1}{1 + (p/K_D)^n}. \quad (43)$$

In this model, Her1 protein decreases the rate at which the promoter switches from OFF to ON. For simplicity, we set $k_{+,0} = 0$. Also for simplicity, we ignore the time delay in mRNA production, since it is not required to generate quasi-periodic solutions (i.e., we set the time delay $\tau = 0$).

Duration regulation:

We model duration regulation via

$$k_{off} = k_{-,0} + k_{-,1} \frac{p^n}{K_D^n + p^n}. \quad (44)$$

In this model, Her1 protein increases the rate at which the promoter switches from ON to OFF. For simplicity, we set $k_{-,0} = 0$. Also for simplicity, we ignore the time delay in mRNA production, since it is not required to generate quasi-periodic solutions (i.e., we set the time delay $\tau = 0$).

3.2 Connection to deterministic models of the segmentation clock

The standard approach to modeling segmentation clock oscillations [3], which predicts smooth mRNA oscillations, is to use a set of delay differential equations describing mRNA and protein concentrations, m and p , that is of the form

$$\dot{m} = r_m f(p(t - \tau)) - \gamma_m m \quad (45)$$

$$\dot{p} = r_p m - \gamma_p p. \quad (46)$$

Here, r_m and r_p are the transcription and translation rates of mRNA and protein, respectively, and γ_m and γ_p are the decay rates of mRNA and protein, respectively. The function $f(p)$ describes auto-regulation, with the rate of transcription modulated by protein concentration. This function typically takes the form of a decreasing Hill function,

$$f(p) = \frac{1}{1 + (p/K_D)^n}. \quad (47)$$

The time delay in the repression term is required mathematically for the system to contain a Hopf bifurcation [3, 4]. Past the Hopf bifurcation, this dynamical system exhibits oscillatory solutions, with a period set by the sum of the delay and the molecular half-lives [3, 4].

We can obtain this deterministic model as a particular limit of our stochastic bursting model. The first step to obtain the deterministic limit is to consider the amplitude regulation scheme in the limit where transcription and translation rates are high, such that small number fluctuations in mRNA and protein concentrations are negligible and we can approximate them by continuous variables that evolve deterministically. In that limit, the system becomes a piece-wise deterministic process,

$$\dot{m} = r_m \chi_t \frac{1}{1 + \left(\frac{p(t-\tau)}{K_D} \right)^n} - \gamma_m m \quad (48)$$

$$\dot{p} = r_p m - \gamma_p p, \quad (49)$$

with χ_t the promoter state variable that stochastically switches between ON and OFF states with constant probability rates k_{on} and k_{off} . Then, we simply take the limit of an always on promoter, $k_{on}/k_{off} \rightarrow \infty$, such that the probability that $\chi_t = 1$ limits to unity. In this limit, we recover the deterministic model.

3.3 Connection to other work

Our model contains as special cases models from various other works. In particular, Wang et al.[5] studied a protein-only version of the duration regulation model in the case of linear repression, $k_{off} = k_{-,1}p$, and found approximate analytic conditions for quasi-regular oscillations. However, the protein oscillations in this linear model are not as regular as experimentally-measured protein oscillations; we found that non-linearity appears to help achieve cleaner oscillations.

Work by Zinani et al. [6] considers a fully stochastic amplitude regulation model, using a high burst frequency of approximately 4 / minute. As we imaged every 1 minute in our experiments, we cannot rule in or out dynamics at that fast of a timescale.

The model of Lengyel et al. [7] does not assume bursting in the absence of auto-repression as our model allows, but posits a mechanism similar to our version of amplitude regulation. By explicitly modeling the binding of Her1 proteins to the *her1* regulatory DNA, they demonstrated that regular oscillations can arise without incorporating additional delays into the system.

The model of Jenkins et al. [8] considers explicit binding and unbinding of Her1/7 dimers to a single binding site and two copies of each gene within an more complex gene regulatory

network. When the model is restricted to only one Her species, it resembles our version of duration regulation, with unbounded regulation of the promoter off rate, $k_{off} \sim p^2$.

Finally, there is a large literature on stochastic oscillations in the circadian clock, including Bratsun et al. [2], which is a more restricted version of our model.

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